

Quantile Regression

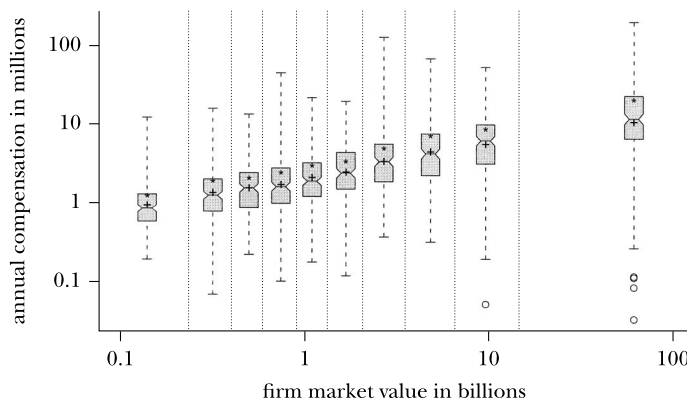
Roger Koenker and Kevin F. Hallock

We say that a student scores at the τ th quantile of a standardized exam if he performs better than the proportion τ of the reference group of students and worse than the proportion $(1-\tau)$. Thus, half of students perform better than the median student and half perform worse. Similarly, the quartiles divide the population into four segments with equal proportions of the reference population in each segment. The quintiles divide the population into five parts; the deciles into ten parts. The quantiles, or percentiles, or occasionally fractiles, refer to the general case. Quantile regression as introduced by Koenker and Bassett (1978) seeks to extend these ideas to the estimation of *conditional quantile functions*—models in which quantiles of the conditional distribution of the response variable are expressed as functions of observed covariates.

In Figure 1, we illustrate one approach to this task based on Tukey's boxplot (as in McGill, Tukey and Larsen, 1978). Annual compensation for the chief executive officer (CEO) is plotted as a function of firm's market value of equity. A sample of 1,660 firms was split into ten groups of equal size according to their market capitalization. For each group of 166 firms, we compute the three quartiles of CEO compensation: salary, bonus and other compensation, including stock options (as valued by the Black-Scholes formula at the time of the grant). For each group, the bow-tie-like box represents the middle half of the salary distribution lying between the first and third quartiles. The horizontal line near the middle of each box represents the median compensation for each group of CEOs, and the

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Figure 1

Pay of Chief Executive Officers by Firm Size

Notes: The boxplots provide a summary of the distribution of CEO annual compensation for ten groupings of firms ranked by market capitalization. The light gray vertical lines demarcate the deciles of the firm size groupings. The upper and lower limits of the boxes represent the first and third quartiles of pay. The median for each group is represented by the horizontal bar in the middle of each box.

Source: Data on CEO annual compensation from EXECUCOMP in 1999.

notches represent an estimated confidence interval for each median estimate. The full range of the observed salaries in each group is represented by the horizontal bars at the end of the dashed “whiskers.” In cases where the whiskers would extend more than three times the interquartile range, they are truncated and the remaining outlying points are indicated by open circles. The mean compensation for each group is also plotted: the geometric mean as a + and the arithmetic mean as a *.

There is a clear tendency for compensation to rise with firm size, but one can also discern several other features from the plot. Even on the log scale, there is a tendency for dispersion, as measured by the interquartile range of log compensation, to increase with firm size. This effect is accentuated if we consider the upper and lower tails of the salary distribution. By characterizing the entire distribution of annual compensation for each group, the plot provides a much more complete picture than would be offered by simply plotting the group means or medians. Here we have the luxury of a moderately large sample size in each group. Had we had several covariates, grouping observations into homogeneous cells, each with a sufficiently large number of observations, would become increasingly difficult.

In classical linear regression, we also abandon the idea of estimating separate means for grouped data as in Figure 1, and we *assume* that these means fall on a line or some linear surface, and we estimate instead the parameters of this linear model. Least squares estimation provides a convenient method of estimating such condi-

tional mean models. Quantile regression provides an equally convenient method for estimating models for conditional quantile functions.

Quantiles via Optimization

Quantiles seem inseparably linked to the operations of ordering and sorting the sample observations that are usually used to define them. So it comes as a mild surprise to observe that we can define the quantiles through a simple alternative expedient as an optimization problem. Just as we can define the sample mean as the solution to the problem of minimizing a sum of squared residuals, we can define the median as the solution to the problem of minimizing a sum of absolute residuals. The symmetry of the piecewise linear absolute value function implies that the minimization of the sum of absolute residuals must equate the number of positive and negative residuals, thus assuring that there are the same number of observations above and below the median.

What about the other quantiles? Since the symmetry of the absolute value yields the median, perhaps minimizing a sum of *asymmetrically* weighted absolute residuals—simply giving differing weights to positive and negative residuals—would yield the quantiles. This is indeed the case. Solving

$$\min_{\xi \in \Re} \sum \rho_{\tau}(y_i - \xi),$$

where the function $\rho_{\tau}(\cdot)$ is the tilted absolute value function appearing in Figure 2 that yields the τ th sample quantile as its solution.¹

Having succeeded in defining the unconditional quantiles as an optimization problem, it is easy to define conditional quantiles in an analogous fashion. Least squares regression offers a model for how to proceed. If, presented with a random sample $\{y_1, y_2, \dots, y_n\}$, we solve

$$\min_{\mu \in \Re} \sum_{i=1}^n (y_i - \mu)^2,$$

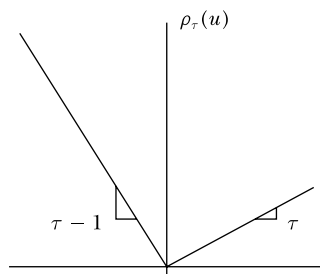
we obtain the sample mean, an estimate of the *unconditional* population mean, EY . If we now replace the scalar μ by a parametric function $\mu(x, \beta)$ and solve

$$\min_{\beta \in \Re^p} \sum_{i=1}^n (y_i - \mu(x_i, \beta))^2,$$

we obtain an estimate of the *conditional* expectation function $E(Y|x)$.

¹ To see this more formally, one need only compute directional derivatives with respect to ξ .

Figure 2

Quantile Regression ρ Function

In quantile regression, we proceed in exactly the same way. To obtain an estimate of the conditional median function, we simply replace the scalar ξ in the first equation by the parametric function $\xi(x_i, \beta)$ and set τ to $\frac{1}{2}$. Variants of this idea were proposed in the mid-eighteenth century by Boscovich and subsequently investigated by Laplace and Edgeworth, among others. To obtain estimates of the other conditional quantile functions, we replace absolute values by $\rho_\tau(\cdot)$ and solve

$$\min_{\beta \in \mathbb{R}^p} \sum \rho_\tau(y_i - \xi(x_i, \beta)).$$

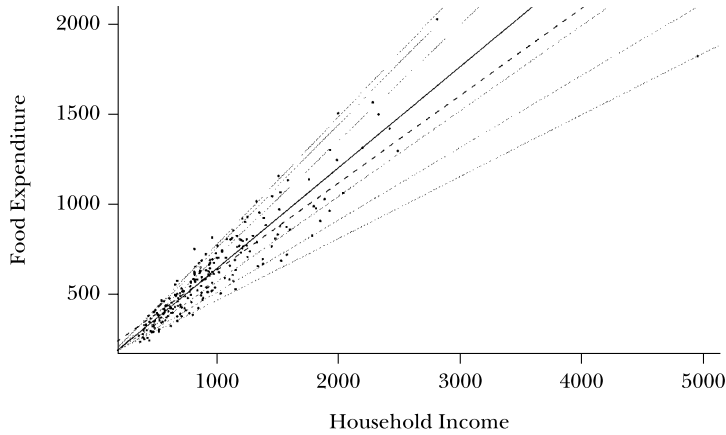
The resulting minimization problem, when $\xi(x, \beta)$ is formulated as a linear function of parameters, can be solved very efficiently by linear programming methods.

Quantile Engel Curves

To illustrate the basic ideas, we briefly reconsider a classical empirical application in economics, Engel's (1857) analysis of the relationship between household food expenditure and household income. In Figure 3, we plot Engel's data taken from 235 European working-class households. Superimposed on the plot are seven estimated quantile regression lines corresponding to the quantiles $\{0.05, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95\}$. The median $\tau = 0.5$ fit is indicated by the darker solid line; the least squares estimate of the conditional mean function is plotted as the dashed line.

The plot clearly reveals the tendency of the *dispersion* of food expenditure to increase along with its level as household income increases. The spacing of the quantile regression lines also reveals that the conditional distribution of food expenditure is skewed to the left: the narrower spacing of the upper quantiles indicating high density and a short upper tail and the wider spacing of the lower quantiles indicating a lower density and longer lower tail.

Figure 3
Engel Curves for Food



Notes: This figure plots data taken from Engel's (1857) study of the dependence of households' food expenditure on household income. Seven estimated quantile regression lines for different values of τ $\{0.05, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95\}$ are superimposed on the scatterplot. The median $\tau = 0.5$ is indicated by the darker solid line; the least squares estimate of the conditional mean function is indicated by the dashed line.

The conditional median and mean fits are quite different in this example, a fact that is partially explained by the asymmetry of the conditional density and partially by the strong effect exerted on the least squares fit by the two unusual points with high income and low food expenditure. Note that one consequence of this nonrobustness is that the least squares fit provides a rather poor estimate of the conditional mean for the poorest households in the sample. Note that the dashed least squares line passes above *all* of the very low income observations.

We have occasionally encountered the faulty notion that something like quantile regression could be achieved by segmenting the response variable into subsets according to its unconditional distribution and then doing least squares fitting on these subsets. Clearly, this form of "truncation on the dependent variable" would yield disastrous results in the present example. In general, such strategies are doomed to failure for all the reasons so carefully laid out in Heckman's (1979) work on sample selection.

In contrast, segmenting the sample into subsets defined according to the conditioning covariates is always a valid option. Indeed, such local fitting underlies all nonparametric quantile regression approaches. In the most extreme cases, we have p distinct cells corresponding to different settings of the covariate vector, x , and quantile regression reduces simply to computing ordinary univariate quantiles for each of these cells. In intermediate cases, we may wish to project these cell estimates onto a more parsimonious (linear) model; see Chamberlain (1994) and Knight, Bassett and Tam (2000) for examples of this approach. Another variant is

the suggestion that instead of estimating linear conditional quantile models, we could instead estimate a family of binary response models for the probability that the response variable exceeded some prespecified cutoff values.²

Quantile Regression and Determinants of Infant Birthweight

In this section, we reconsider a recent investigation by Abrevaya (2001) of the impact of various demographic characteristics and maternal behavior on the birthweight of infants born in the United States. Low birthweight is known to be associated with a wide range of subsequent health problems and has even been linked to educational attainment and eventual labor market outcomes. Consequently, there has been considerable interest in factors influencing birthweights and public policy initiatives that might prove effective in reducing the incidence of low-birthweight infants, which is conventionally defined by whether infants weigh less than 2500 grams at birth, about 5 pounds, 9 ounces.

Most of the analysis of birthweights has employed conventional least squares regression methods. However, it has been recognized that the resulting estimates of various effects on the conditional mean of birthweights were not necessarily indicative of the size and nature of these effects on the lower tail of the birthweight distribution. A more complete picture of covariate effects can be provided by estimating a family of conditional quantile functions.³

Our analysis is based on the June 1997 Detailed Natality Data published by the National Center for Health Statistics. Like Abrevaya (2001), we limit the sample to live, singleton births, with mothers recorded as either black or white, between the ages of 18 and 45, residing in the United States. Observations with missing data for any of the variables described below were dropped from the analysis. This process yielded a sample of 198,377 babies. Birthweight, the response variable, is recorded in grams. Education of the mother is divided into four categories: less than high school, high school, some college and college graduate. The omitted category is less than high school so coefficients may be interpreted relative to this category. The prenatal medical care of the mother is also divided into four categories: those with no prenatal visit, those whose first prenatal visit was in the first trimester of the pregnancy, those with first visit in the second trimester and those with first visit in the last trimester. The omitted category is the group with a first visit in the first trimester; they constitute almost 85 percent of the sample. An indicator of whether

² This approach replaces the hypothesis of conditional quantile functions that are linear in parameters with the hypothesis that some transformation of the various probabilities of exceeding the chosen cutoffs, say the logistic, could instead be expressed as linear functions in the observed covariates. In our view, the conditional quantile assumption is more natural, if only because it nests within it the independent and identically distributed error location shift model of classical linear regression.

³ In an effort to focus attention more directly on the lower tail, several studies have recently explored binary response (probit) models for the occurrence of low birthweights.

the mother smoked during pregnancy is included in the model, as well as mother's reported average number of cigarettes smoked per day. The mother's reported weight gain during pregnancy (in pounds) is included (as a quadratic effect).

Figure 4 presents a summary of quantile regression results for this example. Unlike the Engel curve example, where we had only one covariate and the entire empirical analysis could be easily superimposed on the bivariate scatterplot of the observations, we now have 15 covariates, plus an intercept. For each of the 16 coefficients, we plot the 19 distinct quantile regression estimates for τ ranging from 0.05 to 0.95 as the solid curve with filled dots. For each covariate, these point estimates may be interpreted as the impact of a one-unit change of the covariate on birthweight holding other covariates fixed. Thus, each of the plots has a horizontal quantile, or τ , scale, and the vertical scale in grams indicates the covariate effect. The dashed line in each figure shows the ordinary least squares estimate of the conditional mean effect. The two dotted lines represent conventional 90 percent confidence intervals for the least squares estimate. The shaded gray area depicts a 90 percent pointwise confidence band for the quantile regression estimates.

In the first panel of the figure, the intercept of the model may be interpreted as the estimated conditional quantile function of the birthweight distribution of a girl born to an unmarried, white mother with less than a high school education, who is 27 years old and had a weight gain of 30 pounds, didn't smoke, and had her first prenatal visit in the first trimester of the pregnancy. The mother's age and weight gain are chosen to reflect the means of these variables in the sample.⁴ Note that the $\tau = 0.05$ quantile of the distribution for this group is just at the margin of the conventional definition of a low birthweight baby.

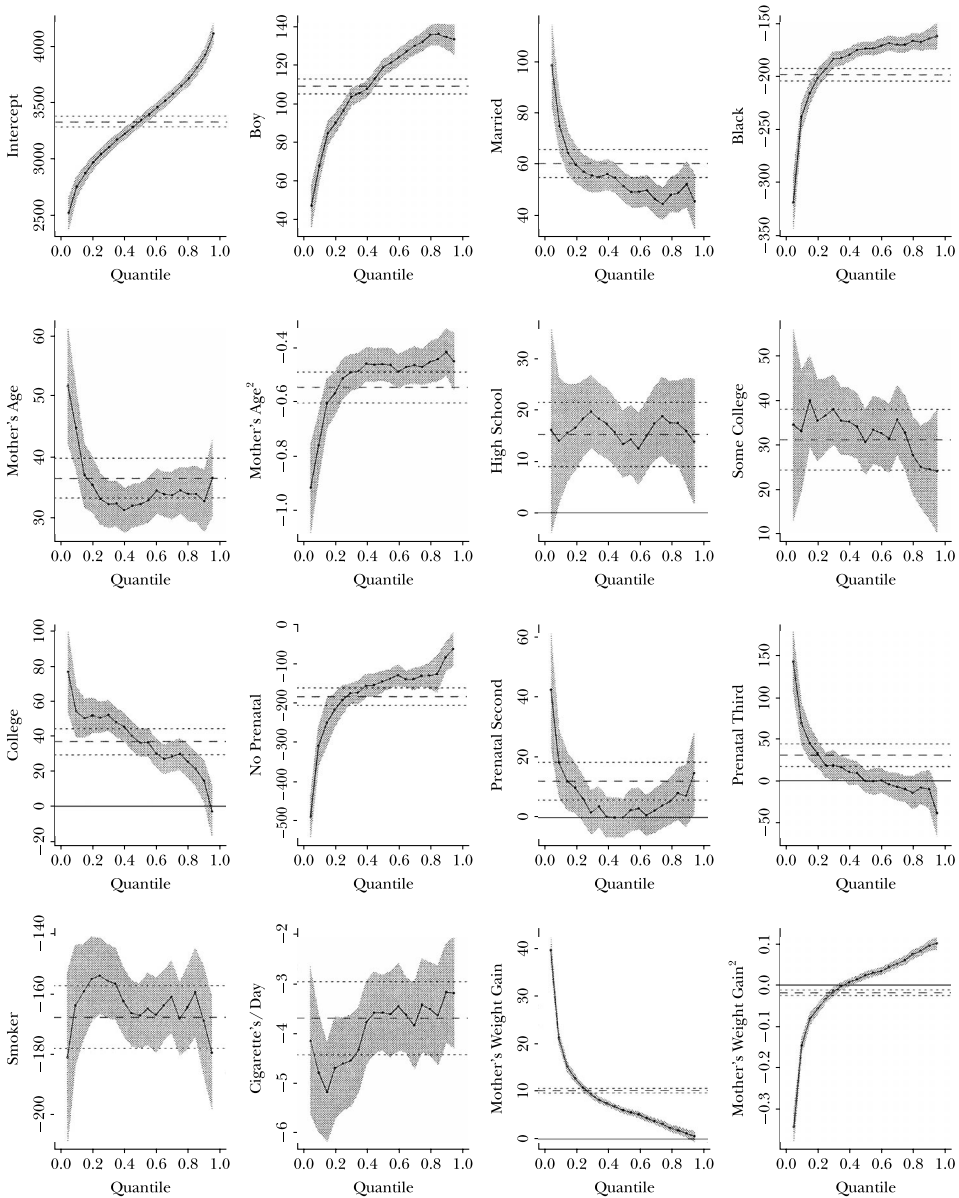
We will confine our discussion to only a few of the covariates. At any chosen quantile we can ask, for example, how different are the corresponding weights of boys and girls, given a specification of the other conditioning variables. The second panel answers this question. Boys are obviously larger than girls, by about 100 grams according to the ordinary least squares estimates of the mean effect, but as is clear from the quantile regression results, the disparity is much smaller in the lower quantiles of the distribution and considerably larger than 100 grams in the upper tail of the distribution. For example, boys are about 45 grams larger at the 0.05 quantile but are about 130 grams larger at the 0.95 quantile. The conventional least squares confidence interval does a poor job of representing this range of disparities.

The disparity between birthweights of infants born to black and white mothers is substantial, particularly at the left tail of the distribution. At the 5th percentile of the conditional distribution, the difference is roughly one-third of a kilogram.

⁴ It is conducive for interpretation to center covariates so that the intercept can be interpreted as the conditional quantile function for some representative case—rather than as an extrapolation of the model beyond the convex hull of the data. This may be viewed as adhering to John Tukey's dictum: "Never estimate intercepts, always estimate *centercepts*!"

Figure 4

Ordinary Least Squares and Quantile Regression Estimates for Birthweight Model



Mother's age enters the model as a quadratic effect, shown in the first two figures of the second row. At the lower quantiles, the mother's age tends to be more concave, increasing birthweight from age 18 to about age 30, but tending to decrease birthweight when the mother's age is beyond 30. At higher quantiles, this

“optimal age” becomes gradually older. At the third quantile, it is about 36, and at $\tau = 0.9$, it is almost 40.

Education beyond high school is associated with a modest increase in birthweights. High school graduation has a quite uniform effect over the whole range of the distribution of about 15 grams. This is a rare example of an effect that really does appear to exert a pure location shift effect on the conditional distribution. For this effect, the quantile regression results are quite consistent with the least squares results; but this is the exceptional case, not the rule.

Several of the remaining covariates are of substantial public policy interest. These include the effects of prenatal care, marital status and smoking. However, as in the corresponding least squares analysis, the interpretation of their causal effects may be somewhat controversial. For example, although we find (as expected) that babies born to mothers with no prenatal care are smaller, we also find that babies born to mothers who delayed prenatal visits until the second or third trimester have substantially *higher* birthweights in the lower tail than mothers who had a prenatal visit in the first trimester. This might be interpreted as the self-selection effect of mothers confident about favorable outcomes.

In almost all of the panels of Figure 4, with the exception of education coefficients, the quantile regression estimates lie at some point outside the confidence intervals for the ordinary least squares regression, suggesting that the effects of these covariates may not be constant across the conditional distribution of the independent variable. Formal testing of this hypothesis is discussed in Koenker and Machado (1999).

Selected Empirical Examples of Quantile Regression

There is a rapidly expanding empirical quantile regression literature in economics that, taken as a whole, makes a persuasive case for the value of “going beyond models for the conditional mean” in empirical economics. Catalyzed by Gary Chamberlain’s (1994) invited address to the 1990 World Congress of the Econometric Society, there has been considerable work in labor economics: on union wage effects, returns to education and labor market discrimination. Chamberlain finds, for example, that for manufacturing workers, the union wage premium at the first decile is 28 percent and declines monotonically to a negligible 0.3 percent at the upper decile. The least squares estimate of the mean union premium of 15.8 percent is thus captured mainly by the lower tail of the conditional distribution. The conventional location shift model thus delivers a rather misleading impression of the union effect. Other contributions exploring these issues in the U.S. labor market include the influential work of Buchinsky (1994, 1997). Arias, Hallock and Sosa-Escudero (2001), using data on identical twins, interpret observed heterogeneity in the estimated returns to education over quantiles as

indicative of an interaction between observed educational attainment and unobserved ability.

There is also a large literature dealing with related issues in labor markets outside the United States, including Fitzenberger (1999) on Germany; Machado and Mata (1999) on Portugal; Garcia, Hernandez and Lopez-Nicolas (2001) on Spain; Schultz and Mwabu (1998) on South Africa; and Kahn (1998) on international comparisons. The work of Machado and Mata (1999) is particularly notable since it introduces a useful way to extend the counterfactual wage decomposition approach of Oaxaca (1973) to quantile regression and provides a general strategy for simulating marginal distributions from the quantile regression process. Tannuri (2000) has employed this approach in a recent study of assimilation of U.S. immigrants.

In other applied micro areas, Eide and Showalter (1998), Knight, Bassett and Tam (2000) and Levin (2001) have addressed school quality issues. Poterba and Rueben (1995) and Mueller (2000) study public-private wage differentials in the United States and Canada. Abadie, Angrist and Imbens (2001) consider estimation of endogenous treatment effects in program evaluation, and Koenker and Billias (2001) explore quantile regression models for unemployment duration data. Work by Viscusi and Hamilton (1999) considers public decision making regarding hazardous waste cleanup.

Deaton (1997) offers a nice introduction to quantile regression for demand analysis. In a study of Engel curves for food expenditure in Pakistan, he finds that although the median Engel elasticity of 0.906 is similar to the ordinary least squares estimate of 0.909, the coefficient at the tenth quantile is 0.879 and the estimate at the 90th percentile is 0.946, indicating a pattern of heteroskedasticity like that of our Figure 3.

In another demand application, Manning, Blumberg and Moulton (1995) study demand for alcohol using survey data from the National Health Interview Study and find considerable heterogeneity in the price and income elasticities over the full range of the conditional distribution. Hendricks and Koenker (1992) investigate demand for electricity by time of day using a hierarchical model.

Earnings inequality and mobility is a natural arena of applications for quantile regression. Conley and Galenson (1998) explore wealth accumulation in several U.S. cities in the mid-nineteenth century. Gosling, Machin and Meghir (2000) study the income and wealth distribution in the United Kingdom. Trede (1998) and Morillo (2000) compare earnings mobility in the United States and Germany.

There is also a growing literature in empirical finance employing quantile regression methods. One strand of this literature is the rapidly expanding literature on value at risk: this connection is developed in Taylor (1999), Chernozhukov and Umantsev (2001) and Engle and Manganelli (1999). Bassett and Chen (2001) consider quantile regression index models to characterize mutual fund investment styles.

Software and Standard Errors

The diffusion of technological change throughout statistics is closely tied to its embodiment in statistical software. This is particularly true of quantile regression methods, since the linear programming algorithms that underlie reliable implementations of the methods appear somewhat esoteric to some users. Since the early 1950s, it has been recognized that median regression methods based on minimizing sums of absolute residuals can be formulated as linear programming problems and efficiently solved with some form of the simplex algorithm.⁵

Among commercial programs in common use in econometrics, Stata and TSP offer some basic functionality for quantile regression within the central core of the package distributed by the vendor. Since the mid-1980s, one of us has maintained a public domain package of quantile regression software designed for the *S* language of Becker, Chambers and Wilks (1988) and the related commercial package Splus. Recently, this package has been extended to provide a version for *R*, the impressive public domain dialect of *S* (Koenker, 1995). This website also provides quantile regression software for Ox and Matlab languages.

It is a basic principle of sound econometrics that *every serious estimate deserves a reliable assessment of precision*. There is now an extensive literature on the asymptotic behavior of quantile regression estimators and considerable experience with inference methods based on this theory, as well as a variety of resampling schemes. We have recently undertaken a comparison of several approaches to the construction of confidence intervals for a problem typical of current applications in labor economics. We find that the discrepancies among competing methods are slight, and inference for quantile regression is, if anything, more robust than for most other forms of inference commonly encountered in econometrics. Koenker and Hallock (2000) describe this exercise in more detail and provide a brief survey of recent work on quantile regression for discrete data models, time series, nonparametric models and a variety of other areas. Some of these developments have been slow to percolate into standard econometric software. With the notable exceptions of Stata and Xplore (Cizek, 2000) and the packages available from the web for Splus and *R*, none of the implementations of quantile regression in econometric software packages include any functionality for inference.

⁵ The median regression algorithm of Barrodale and Roberts (1974) has proven particularly influential and can be easily adapted for general quantile regression. Koenker and d'Orey (1987) describe one implementation. For large-scale quantile regression problems, Portnoy and Koenker (1997) have shown that a combination of interior point methods and effective preprocessing renders quantile regression computation competitive with least squares computations for problems of comparable size.

Conclusion

Much of applied statistics may be viewed as an elaboration of the linear regression model and associated estimation methods of least squares. In beginning to describe these techniques, Mosteller and Tukey (1977, p. 266) remark in their influential text:

What the regression curve does is give a grand summary for the averages of the distributions corresponding to the set of x 's. We could go further and compute several different regression curves corresponding to the various percentage points of the distributions and thus get a more complete picture of the set. Ordinarily this is not done, and so regression often gives a rather incomplete picture. Just as the mean gives an incomplete picture of a single distribution, so the regression curve gives a corresponding incomplete picture for a set of distributions.

We would like to think that quantile regression is gradually developing into a comprehensive strategy for completing the regression picture.

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REGRESSION QUANTILES¹

BY ROGER KOENKER AND GILBERT BASSETT, JR.

A simple minimization problem yielding the ordinary sample quantiles in the location model is shown to generalize naturally to the linear model generating a new class of statistics we term "regression quantiles." The estimator which minimizes the sum of absolute residuals is an important special case. Some equivariance properties and the joint asymptotic distribution of regression quantiles are established. These results permit a natural generalization to the linear model of certain well-known robust estimators of location.

Estimators are suggested, which have comparable efficiency to least squares for Gaussian linear models while substantially out-performing the least-squares estimator over a wide class of non-Gaussian error distributions.

1. INTRODUCTION

IN STATISTICAL PARLANCE the term robustness has come to connote a certain resilience of statistical procedures to deviations from the assumptions of hypothetical models. The paradigm may be briefly stated as follows.² The process generating observed data is thought to be approximately described by an element of some parametric class of models. The investigator seeks statistics, i.e., a mapping from the sample space to a parameter space, whose distribution will be as concentrated as possible near the true parameters—if the hypothesized model is correct. If however, as seems almost certain, the parametric model is not quite true, one would like to use estimators whose distributions were altered only slightly if the distribution of the observations were close, in some reasonable sense, to that of some member of the parametric class. In important special cases this modest robustness requirement is not met by estimators in common use.³

We consider the familiar problem of estimating a vector of unknown (regression) parameters, β , from a sample of independent observations on random variables $Y_1, Y_2, \dots, Y_T$, distributed according to

$$(1.1) \quad P(Y_t < y) = F(y - x_t \beta) \quad (t = 1, \dots, T),$$

where $x_t: t = 1, \dots, T$, denote rows of a known $(T \times K)$ design matrix and the shape of F is not precisely known.⁴ If F is known precisely then it is frequently

¹ An early version of this paper [5] was presented at the Winter, 1974 meeting of the Econometric Society in San Francisco.

² A rigorous statement of this point of view on the robustness problem may be found in the work of Hampel [16, 17, 18].

³ We need only mention the sample mean, an estimator *nonpareil* of location if the sample observations are generated by an independent and identically distributed Gaussian process. However, in any open neighborhood of a Gaussian distribution there exists distribution functions which would take the distribution of the sample mean arbitrarily far away from its distribution in the Gaussian case; cf., Hampel [16].

⁴ This formulation restricts attention to what may be called the "robustness to distributional assumptions" problem, and leaves aside problems involving possible dependence among observations, non-linearities of functional form, etc.

possible to show that the maximum likelihood estimator or some one-step (M) approximant to it is efficient in the Cramér-Rao sense. In particular, when F is known to be Gaussian (normal), Rao has shown that the least squares estimator, $\hat{\beta}$, is minimum variance in the class of unbiased estimators. Unfortunately the extreme sensitivity of the least squares estimator to modest amounts of outlier contamination makes it a very poor estimator in many non-Gaussian, especially long-tailed, situations. This paper introduces new classes of robust alternatives to the least squares estimator for the linear model. Estimators are suggested which have comparable efficiency to $\hat{\beta}$ for Gaussian models while substantially outperforming the least squares estimator over a wide class of non-Gaussian error distributions. The proposed estimators are analogues to linear combinations of sample quantiles in the location model.⁵

2. BACKGROUND AND MOTIVATION

The aphorism made famous by Poincaré and quoted by Cramér [12] that, “everyone believes in the [Gaussian] law of errors, the experimenters because they think it is a mathematical theorem, the mathematicians because they think it is an experimental fact,” is still all too apt. This “dogma of normality” as Huber has called it, seems largely attributable to a kind of wishful thinking. As Box and Tiao put it,

Classical statistical arguments lead us to treat assumptions as if they were in some way axiomatic and yet consideration will show that, in fact, they are conjectures which in practice may be expected to be more or less true If we assume normality, we can proceed with an ‘objective’ classical analysis . . . however . . . as seems to be inevitably the case in other problems as well as this one, our ‘objectivity’ is gained by pretending to knowledge we do not have . . . [11, p. 419].

Following Haavelmo’s [15] classic paper it is sometimes argued that the errors encountered in econometric models are known to be the sum of a large number of small and independent elementary errors, and therefore will be approximately Gaussian due to central limit theorem considerations. However, it is rather puzzling that investigators, who are generally loathe to adopt informative priors about the systematic structure of their models about which theoretical considerations and past empirical experience should provide substantive evidence, should feel themselves so well informed about the unobservable constituents of their model’s unobservable errors to argue that they satisfy a Lindeberg condition! A few gross errors occurring with low probability can cause serious deviations from normality: to dismiss the possibility of these occurrences almost invariably requires a leap of Gaussian faith into the realm of pure speculation.

The need for robust alternatives to the sample mean (the least squares estimator in the location model) has been apparent since the eighteenth century. The median, other trimmed means, and more complicated linear combinations of order statistics were in common use especially in astronomical calculations, in the

⁵ For obvious reasons the term “location model” is used to describe (1.1) when $x_t = 1: t = 1, \dots, T$.

nineteenth century.⁶ By 1821 Gauss had shown that the sample mean provided the “most probable” estimate of the location parameter from a random sample with probability density proportional to $e^{-x^2/2\sigma^2}$, but this result was *explicitly an ex post rationalization* for the use of the sample mean rather than a claim for the empirical validity of this particular error law.⁷ In fact, it was noted by a number of authors that error distributions with longer tails than that of the Gaussian distribution were commonly observed. In such cases it appeared desirable to choose estimators which modified the sample mean by putting reduced weight on extreme observations.

There was a parallel early recognition of the need for robust alternatives to the least squares estimator for the linear model. Wild observations, or “outliers” as they came to be called, were more difficult to identify in such models and the fruitful notion from the location model of an ordering of sample observations had no simple analogue in the more complicated models. Many illustrious figures (Gauss, Laplace and Legendre, to name only three) suggested that the minimization of absolute deviations might be preferable to least squares when some sample observations are of dubious reliability.⁸ In 1818 Laplace proved that in the simple model of bivariate regression through the origin, this least absolute error (LAE) estimator had smaller asymptotic variance than the least squares estimator if the error law of the model had variance, σ^2 , and density at the median, $f(0)$, satisfying $[2f(0)]^{-1} < \sigma$. This result paved the way for investigations of the large sample theory of statistics based on sample quantiles in the location model.⁹

The elementary point that there may exist nonlinear, or for that matter—biased, estimators superior to least squares for the non-Gaussian linear model is a well kept secret in most of the econometrics literature. Statements of the Gauss-Markov theorem too often seem to imply that linearity in y and unbiasedness are added virtues of the least squares estimator instead of restrictions on the class of its potential competitors. Indeed one sometimes encounters the view that infinite variance of the errors constitutes the only possible rationale for seeking robust alternatives to least squares in the linear model. This is, emphatically, false. While least squares is obviously abysmal for distributions having infinite variance (having zero efficiency for the Cauchy for example) its gross inferiority to a variety of nonlinear estimators is by no means confined to distributions with infinite variance.

The wave of current interest¹⁰ in the problem of robust estimation has focused primarily on the location model. While we cannot hope to do justice to the vast recent literature on this subject we briefly sketch the main lines of the developments which are most relevant to our work on the linear model.

⁶ See the excellent survey of Harter [19] and the fascinating paper by Stigler [38].

⁷ This point is made emphatically by Huber [25] in his survey paper on robust estimation.

⁸ Boscovitch is generally credited with first proposing estimators which minimize the sum of absolute deviations. See Harter [19].

⁹ Our paper [6] generalizes this result to the general linear model.

¹⁰ Tukey has written of robust estimation as the third wave of statistical theory (see Hampel [18]); after parametric and non-parametric theory, a theory of “almost parametric models” is slowly emerging. See also the fundamental survey papers of Huber [25] and Hogg [21].

Mosteller [32], in 1946, proposed the use of a variety of so-called “inefficient statistics” based on a few sample quantiles as “quick and dirty” substitutes for more conventional estimators. It was found that estimators of this type could be constructed which were almost as efficient as the maximum likelihood estimators for most conventional parametric models. This approach was further developed by Bennett [7] for strictly parametric models, while Gastwirth [14] and others established that some estimators of this type had good efficiency properties for a wide variety of distributions. For example, the weighted average of the $1/3$, $1/2$, and $2/3$ quantiles with weights $.3$, $.4$, $.3$ has asymptotic efficiency of nearly eighty per cent for the Gaussian, Laplace, logistic, and Cauchy distributions.¹¹ In contrast, the sample mean has asymptotic efficiency of one in the Gaussian case, but is less than half as efficient as the median for the Laplace distribution and has zero efficiency for the Cauchy distribution. Thus, although these “quick and dirty” estimators may be “inefficient statistics” for any particular parametric model, in practice they may actually be preferable to putatively “optimal” estimators, like the sample mean, if there is some uncertainty about the shape of the distribution generating the sample. Much recent work has been devoted to extending results of this type beyond a fixed number of quantiles to estimators which are linear combinations of order statistics. The asymptotic theory of such (L) estimators has achieved almost classical standing through the efforts of Bickel [8], Stigler [39] and others. The most common (L) estimator of location is the α -trimmed mean which is simply the mean of the sample after the proportion α of largest and smallest observation have been removed. This venerable¹² estimator was revived by Tukey in the late forties and has played an important role in recent work on robust estimation of location. Huber’s now classic paper [24] on robust estimation of location solves for the least favorable (minimal Fisher information) distribution in the class of Gaussian contaminants—distributions of the form $F = (1 - \varepsilon)\Phi + \varepsilon H$, where Φ denotes the standard Gaussian cumulative, H ranges over all symmetric cumulative distribution functions, and $0 \leq \varepsilon < 1$ is a fixed proportion of contamination: the least favorable distribution has exponential tails, and Gaussian center so the minimax estimator is quadratic in the center and linear in the tails. Asymptotically Huber’s minimax estimator behaves like an ε -trimmed mean.

In order to provide some quantitative evidence on the performance of some alternative estimates of location we have abstracted a small subset of estimators and a small subset of distributions from those considered in the Princeton Robustness Study [3]. Table I gives Monte-Carlo variances for six selected estimators and five selected distributions. We have purposely chosen simple estimators which are nonadaptive. The table clearly illustrates that estimators of location are available which, while making a small sacrifice of efficiency to the mean at the Gaussian distribution, are greatly superior to the mean for non-Gaussian distributions. Only an unshakable *a priori* faith in the Gaussian “law of

¹¹ This estimator is due to Gastwirth [14] and bears his name in the Princeton Robustness Study [3].

¹² Stigler [38] and Harter [19] both discuss the historical background of the trimmed mean.

TABLE I
EMPIRICAL VARIANCES OF SOME ALTERNATIVE LOCATION ESTIMATORS^a
(Sample Size 20)

Estimators	Distributions				
	Normal	10% 3 σ ^b	10% 10 σ ^c	Laplace	Cauchy
Mean	1.00	1.88	11.54	2.10	12,548.0
10% trimmed mean	1.06	1.31	1.46	1.60	7.3
25% trimmed mean	1.20	1.41	1.47	1.33	3.1
Median	1.50	1.70	1.80	1.37	2.9
Gastwirth ^d	1.23	1.45	1.51	1.35	3.1
Trimean ^e	1.15	1.37	1.48	1.43	3.9

^a Abstracted from Exhibit 5 in Andrews, *et al.* [3].

^b Gaussian Mixture: $.9\Phi(1) + .1\Phi(3)$.

^c Gaussian Mixture: $.9\Phi(1) + .1\Phi(10)$.

^d $\hat{\beta} = .3\beta^*(1/3) + .4\beta^*(1/2) + .3\beta^*(2/3)$, where $\beta^*(\theta)$ is the θ th sample quantile.

^e $\hat{\beta} = 1/4\beta^*(1/4) + 1/2\beta^*(1/2) + 1/4\beta^*(3/4)$.

errors” would seem to justify selecting the sample mean.¹³ In practice, of course, it is common to discard certain observations which seem to be deviant on the basis of a preliminary inspection of the data. This procedure obviously amounts to a rough and ready trimmed mean, but more formal procedures seem desirable. Such ad hoc empirical accommodations are even more problematic within the context of the linear model since deviant observations are more difficult to identify there. In Sections 4 through 6 below we propose new classes of robust alternatives to the least squares estimator for the linear model and show that members of these classes have, unlike the least squares estimator, high efficiency over a wide range of error distributions.

Huber’s work on the location model has been extended to other estimators of the (M) maximum likelihood type by Relles [35], Huber [26], Andrews [2], Bickel [10], and others for the linear model. Another line of inquiry, based on analogues of rank procedures in the location model, has been extended to the linear model by Jurečková [29], Jaeckel [28], and others. Bickel [9] has suggested a third line of attack based on analogues of linear combinations of order statistics, (L) estimates, from the linear model. His estimates are one-step iterations from an ordering of observations based on some preliminary robust estimate of location like, for example, the LAE estimate.¹⁴

Our approach, although substantially different from that taken by Bickel, may also be viewed as an attempt to extend to the linear model the notions of systematic statistics and linear combinations of order statistics which have proven so fruitful in dealing with the robust estimation problem in the location model. We begin by introducing a natural generalization to the linear model of the concept of

¹³ Based on the full study of 65 different estimators and ten distributions, Hampel concludes that the sample mean is the “clear candidate for . . . worst estimator of the study.” Andrews, *et al.*, [3, p. 239].

¹⁴ A number of more sophisticated robust (M) estimators use this (LAE) estimator as an initial robust estimate and make a one-step Newton iteration. See, *e.g.*, Hill and Holland [20].

sample quantiles from the location model.¹⁵ The LAE estimator will be an important special case.¹⁶

3. REGRESSION QUANTILES

Our point of departure is an elementary definition of the sample quantiles which, by circumventing the usual reliance on an ordered set of sample observations, is readily extended to the linear model. As above, let $\{y_t: t = 1, \dots, T\}$ be a random sample on a random variable Y having distribution function F . Then the θ th sample quantile, $0 < \theta < 1$, may be defined as any solution to the minimization problem:

$$\min_{b \in \mathbb{R}} \left[\sum_{t \in \{t: y_t \geq b\}} \theta |y_t - b| + \sum_{t \in \{t: y_t < b\}} (1 - \theta) |y_t - b| \right].$$

The case of the median ($\theta = 1/2$) is, of course, well known, but the general result has languished in the status of curiosum—appearing for example as an exercise in Ferguson [13].

Huber's [26] observation that outliers are difficult to identify in the regression context underlines the essential ambiguity involved in extending to the linear model the ordinary notions of sample quantiles based on an ordering of sample observations. A direct generalization of the minimization problem posed above resolves this ambiguity. Letting $\{x_t: t = 1, \dots, T\}$ denote a sequence of (row) K -vectors of a known design matrix, suppose $\{y_t: t = 1, \dots, T\}$ is a random sample on the regression process $u_t = y_t - x_t\beta$ having distribution function F . The θ th regression quantile, $0 < \theta < 1$, is defined as any solution to the minimization problem:

$$\min_{b \in \mathbb{R}^K} \left[\sum_{t \in \{t: y_t \geq x_t b\}} \theta |y_t - x_t b| + \sum_{t \in \{t: y_t < x_t b\}} (1 - \theta) |y_t - x_t b| \right].$$

In the location model ($K = 1$, $x_t = 1$, for all t) the two minimization problems coincide. The least absolute error estimator is the regression median, i.e., the regression quantile for $\theta = 1/2$.

We now introduce some crucial notation and state some fundamental properties of elements, $\beta^*(\theta)$, of the solution sets $B^*(\theta)$ of the regression quantile minimization problem.

Let $\mathcal{T} = \{1, 2, \dots, T\}$ and $\mathcal{H}$ denote the set of K -element subsets of $\mathcal{T}$. Elements $h \in \mathcal{H}$ have relative complement, $\bar{h} = \mathcal{T} - h$, and both serve to partition

¹⁵ Professor Hogg [22, 23] has recently proposed an alternative method of estimating "percentile hyperplanes" for the linear model which revives and generalizes the median regression methods of Mood and Brown. While the details of his method differ greatly from ours, the approaches are quite similar in spirit.

¹⁶ The large sample theory of LAE is discussed in [6]; see also Taylor [41] for a recent survey, and Bassett [4]. Examples of empirical applications where LAE has performed extremely well in comparison with least squares in forecasting tests may be found in Meyer and Glauber [31] and Overson [33].

y and X . Thus, for example $y(h)$ denotes the (K) -vector with elements $\{y_t: t \in h\}$ while $X(h)$ denotes a $(T-K) \times K$ matrix with rows $\{x_t: t \in h\}$. The (K) -dimensional vector of ones will be denoted by ι_K . Finally, let,

$$H = \{h \in \mathcal{H} \mid \text{rank } X(h) = K\}.$$

THEOREM 3.1: *If X has rank K then the set of regression quantiles, $B^*(\theta)$, has at least one element of the form,*

$$\beta^*(\theta) = X(h)^{-1}y(h)$$

for some $h \in H$. Moreover, $B^(\theta)$, is the convex hull of all solutions having this form.*

PROOF: This result follows immediately from the linear programming formulation of the defining minimization problem; see Appendix 1 and the recent paper by Abdelmalek [1] for details.

REMARK: Sample quantiles in the location model are either identified with a single order statistic from the observed sample, or, for example in the case of the median from an even sample, they are identified with a closed interval between two adjacent order statistics. Theorem 1 generalizes this feature to regression quantiles where normals to hyperplanes defined by subsets of K observations play the role of order statistics.

THEOREM 3.2: *If $\beta^*(\theta, y, X) \in B^*(\theta, y, X)$ then the following are elements of the solution of the specified transformed problems:*

- (i) $\beta^*(\theta, \lambda y, X) = \lambda \beta^*(\theta, y, X), \quad \lambda \in [0, \infty),$
- (ii) $\beta^*(1 - \theta, \lambda y, X) = \lambda \beta^*(\theta, y, X), \quad \lambda \in (-\infty, 0],$
- (iii) $\beta^*(\theta, y + X\gamma, X) = \beta^*(\theta, y, X) + \gamma, \quad \gamma \in \mathbb{R}^K,$
- (iv) $\beta^*(\theta, y, XA) = A^{-1}\beta^*(\theta, y, X), \quad A_{K \times K} \text{ nonsingular.}$

PROOF: Let

$$\begin{aligned} \psi(b; \theta, y, X) &= \sum_{\{t: y_t > x_t b\}} \theta |y_t - x_t b| + \sum_{\{t: y_t < x_t b\}} (1 - \theta) |y_t - x_t b| \\ &= \sum_{t=1}^T [\theta - 1/2 + 1/2 \operatorname{sgn}(y_t - x_t b)] [y_t - x_t b] \end{aligned}$$

where $\operatorname{sgn}(u)$ takes values 1, 0, -1 as $u \geq 0$. Now, note that

- (i) $\lambda \psi(b; \theta, y, X) = \psi(\lambda b; \theta, \lambda y, X), \quad \lambda \in [0, \infty),$
- (ii) $-\lambda \psi(b; \theta, y, X) = \psi(\lambda b; 1 - \theta, \lambda y, X), \quad \lambda \in (-\infty, 0],$
- (iii) $\psi(b; \theta, y, X) = \psi(b + \gamma; y + X\gamma, X), \quad \gamma \in \mathbb{R}^K,$
- (iv) $\psi(b; y, X) = \psi(A^{-1}b; y, XA), \quad |A_{K \times K}| \neq 0.$

REMARK: Theorem 3.2 collects a number of equivariance properties of regression quantiles. Note that (i) and (ii) imply $\beta^*(1/2)$ is scale equivariant, (iii) states $\beta^*(\theta)$ is location (or "shift" or "regression") equivariant, and (iv) states that $\beta^*(\theta)$ is equivariant to reparameterization of design.

THEOREM 3.3: *If F is continuous then with probability one: $\beta^* = X(h)^{-1}y(h)$ is a unique solution to Problem (P) if and only if,*

$$(3.1) \quad (\theta - 1)\iota'_K < \sum_{t \in h} [1/2 - 1/2 \operatorname{sgn}(y_t - x_t \beta^*) - \theta] x_t X(h)^{-1} < \theta \iota'_K.$$

PROOF: Consider the directional derivative of $\psi(b)$ in direction w ,

$$(3.2) \quad \psi'(b; w) = \sum_{t=1}^T [1/2 - 1/2 \operatorname{sgn}^*(y_t - x_t b; -x_t w) - \theta] x_t w$$

where

$$\operatorname{sgn}^*(u; z) = \begin{cases} \operatorname{sgn} u & \text{if } u \neq 0, \\ \operatorname{sgn} z & \text{if } z = 0. \end{cases}$$

Since $\psi(b)$ is convex, it attains a unique minimum at β^* if and only if $\psi'(\beta^*; w) > 0$ for all $w \neq 0$. At $\beta^* = X(h)^{-1}y(h)$,

$$(3.3) \quad \begin{aligned} \psi'(\beta^*; w) &= \sum_{t \in h} [1/2 + \operatorname{sgn}(x_t w) - \theta] x_t w \\ &\quad + \sum_{t \in h} [1/2 - 1/2 \operatorname{sgn}^*(y_t - x_t \beta^*; -x_t w) - \theta] x_t w. \end{aligned}$$

Letting $v = X(h)w$, we have that $\psi'(\beta^*; w) > 0$ for all $w \neq 0$, if and only if,

$$(3.4) \quad \begin{aligned} 0 &< \sum_{k=1}^K [(1/2 - \theta)v_k + |v_k|] \\ &\quad + \sum_{t \in h} [1/2 - 1/2 \operatorname{sgn}^*(y_t - x_t \beta^*; x_t X(h)^{-1}v) - \theta] x_t X(h)^{-1}v \end{aligned}$$

for all $v \neq 0$. But this is equivalent to

$$(3.5) \quad \begin{aligned} (\theta - 1)\iota'_K &< \sum_{t \in h} [1/2 - 1/2 \operatorname{sgn}^*(y_t - x_t \beta^*; x_t X(h)^{-1}v) - \theta] x_t X(h)^{-1}v \\ &< \theta \iota'_K, \end{aligned}$$

for all $v \neq 0$. Finally F continuous implies that the events $[y_t - x_t X(h)^{-1}y(h) = 0, t \in h]$ have probability measure zero since they require $M > K$ observations to lie exactly on hyperplanes of dimension $K - 1$. Thus (3.5) simplifies to (3.1).

REMARK: It is perhaps instructive to pause to consider the ordinary sample quantiles in the light of Theorem 3.3. If $x_t = 1$, $t = 1, \dots, T$, so $H = \mathcal{T}$ and F is continuous, then Theorem 3.3 asserts that $\beta^*(\theta) = y(h)$ is a unique θ th sample quantile if and only if,

$$(3.6) \quad \theta - 1 < \sum_{t \in h} [1/2 - 1/2 \operatorname{sgn}(y_t - y(h)) - \theta] < \theta.$$

The expression in brackets takes the values of $-\theta$ if $y_i > y(h)$ and $1 - \theta$ if $y_i < y(h)$, so (3.6) reduces to the requirement that the number of y_i 's less than $y(h)$ be strictly between $T\theta - 1$ and $T\theta$. This in turn demands that $T\theta$ be nonintegral. The continuity of F removes the tiresome problem of "ties" in the location model and accomplishes the same task in the general linear model. It may be noted that in the absence of this degeneracy phenomena the condition for uniqueness is purely a design condition, reducing in the location model to the requirement that $T\theta$ be non-integral. This suggests that for any sequence $\{X_T\}$ of designs one should be able to extract a subsequence, or at worst some "perturbed" subsequence whose elements have unique solutions. An alternative approach which is frequently employed in the location model is to adopt some arbitrary rule to choose a single element from sets of quantiles when they occur. Either approach suffices to obtain the sequence of unique solutions considered in the next section dealing with the large sample distribution of regression quantiles.

THEOREM 3.4: *Let $P(u^*(\theta))$, $N(u^*(\theta))$, and $Z(u^*(\theta))$ denote the number of positive, negative, and zero elements in the vector $u^*(\theta) = y - X\beta^*(\theta)$. Then if X contains a column of ones,*

$$(3.7) \quad N(u^*) \leq T\theta \leq T - P(u^*) = N(u^*) + Z(u^*)$$

for every $\beta^(\theta) \in B^*(\theta)$. If β^* is unique, i.e., $\beta^* = B^*$, then the inequalities hold strictly.*

PROOF: Partition the design so that $X = [\iota : \tilde{X}]$. By the argument used to obtain Theorem 3.3, $\beta^* \in B^*$ if and only if

$$(3.8) \quad \psi'(\beta^*; w) = \sum_{i=1}^T [1/2 - 1/2 \operatorname{sgn}^*(y_i - x_i\beta^*; -x_iw) - \theta]x_iw \geq 0,$$

for all $w \neq 0$. So in particular (3.8) must hold for $w^+ = (1, 0, \dots, 0)$ and $w^- = (-1, 0, \dots, 0)$ in $\mathbb{R}^K$. Since $x_iw^+ = 1$ and $x_iw^- = -1$ for all x_i , (3.8) implies

$$(3.9) \quad \sum_{i=1}^T \pm [1/2 - 1/2 \operatorname{sgn}^*(y_i - x_i\beta^*; \mp 1) - \theta] > 0$$

which is equivalent to the two conditions

$$-\theta P + (1 - \theta)N + (1 - \theta)Z > 0,$$

$$-\theta P + (1 - \theta)N - Z < 0,$$

which in turn are equivalent to (3.7). If $\beta^*(\theta)$ is unique then all inequalities are strict.

REMARK: Note that if F is continuous then $Z(u^*) = K$ with probability one, so there are at least $T\theta$ observations below the θ th regression quantile hyperplane and at most $T\theta + K$ observations above it.

THEOREM 3.5: *If $\beta^*(\theta) \in B^*(\theta, y, X)$, then $\beta^*(\theta) \in B^*(\theta, X\beta^* + Du^*, X)$ where $u^* = y - X\beta^*$ and D is any $T \times T$ diagonal matrix with nonnegative elements.*

PROOF: $\beta^* \in B^*(\theta, y, X)$ implies

$$\sum_{t=1}^T [1/2 - 1/2 \operatorname{sgn}^*(y_t - x_t \beta^*; -x_t w) - \theta] x_t w \geq 0$$

for all $w \neq 0$. Note that

$$\begin{aligned} & [1/2 - 1/2 \operatorname{sgn}^*(x_t \beta^* + d_t(y_t - x_t \beta^*) - x_t \beta^*; -x_t w) - \theta] x_t w \\ &= (1/2 - \theta) x_t w - 1/2 \operatorname{sgn}^*(d_t(y_t - x_t \beta^*); -x_t w) x_t w \\ &\geq (1/2 - \theta) x_t w - 1/2 \operatorname{sgn}^*(y_t - x_t \beta^*; -x_t w) x_t w \end{aligned}$$

for $d_t \geq 0$ and the result follows.

REMARK: This result has a simple geometric interpretation. Imagine a scatter of sample observations in $\mathbb{R}$ with the θ th regression quantile line slicing through the scatter. Now consider the effect (on the position of the θ th *RQ* line) of moving observations up or down in the scatter. The result states that as long as these movements leave observations on the same side of the original line its position is unaffected. This property is obvious in the location model, sample quantile context, but its generalization is perhaps somewhat less obvious.

We conclude this section with an illustration in Table II of the regression quantiles for all $\theta \in (0, 1)$ in a simple bivariate model with five observations. Note that for $\theta \in \{7/22, 1/2, 3/4\}$ the problem admits multiple solutions.

TABLE II

	$0 < \theta \leq 7/22$	$7/22 \leq \theta \leq 1/2$	$1/2 \leq \theta \leq 3/4$	$3/4 \leq \theta < 1$
$\beta_1^*(\theta)$	6/7	21/8	13/6	17/3
$\beta_2^*(\theta)$	4/7	3/8	5/6	1/3

4. THE ASYMPTOTIC DISTRIBUTION THEORY OF REGRESSION QUANTILES

The following well known result concerning sample quantiles in the location model is due to Mosteller [32].

THEOREM 4.1: Let $\{\xi_T^*(\theta_1), \dots, \xi_T^*(\theta_M)\}$ with $0 < \theta_1 < \theta_2 < \dots < \theta_M < 1$, denote a sequence of unique sample quantiles from random samples of size T from a population with inverse distribution function $\xi(\theta) = F^{-1}(\theta)$. If F is continuous and has continuous and positive density, f , at $\xi(\theta_i)$, $i = 1, \dots, M$, then,

$$\sqrt{T}[\xi_T^*(\theta_1) - \xi(\theta_1), \dots, \xi_T^*(\theta_M) - \xi(\theta_M)]$$

converges in distribution to an (M) -variate Gaussian random vector with mean, 0, and covariance matrix $\Omega(\theta_1, \dots, \theta_M; F)$ with typical element,

$$\omega_{ij} = \frac{\theta_i(1 - \theta_j)}{f(\xi(\theta_i))f(\xi(\theta_j))}, \quad i \leq j.$$

This theorem provides the foundation for a large-sample theory of so-called “systematic statistics”—estimators which are linear combinations of a few sample quantiles. The median is the most important special case having asymptotic variance $[2f(\xi(1/2))]^{-2}$. As we have noted above, this value will be less than the variance of the mean for a large class of long-tailed distributions. The asymptotic variance of the Gastwirth or trimean estimator of Table I can be easily calculated in a similar manner from Theorem 4.1 for arbitrary distributions. We merely require the evaluation of the density function of the distribution F at specified quantiles.

The analogy between sample quantiles in the location model and regression quantiles in the linear model is considerably strengthened by the striking resemblance in their asymptotic behavior. This is made explicit in the following theorem which plays a central role in the theory developed in the remainder of the paper.

THEOREM 4.2: *Let $\{\beta_T^*(\theta_1), \beta_T^*(\theta_2), \dots, \beta_T^*(\theta_M)\}$ with $0 < \theta_1 < \theta_2 < \dots < \theta_M < 1$ denote a sequence of unique regression quantiles from model (1.1). Let $\xi(\theta) = F^{-1}(\theta)$, $\xi(\theta) = (\xi(\theta), 0, \dots, 0) \in \mathbb{R}^K$ and $\xi_T^*(\theta) = \beta_T^*(\theta) - \beta$. Assume:*

- (i) *F is continuous and has continuous and positive density, f , at $\xi(\theta_i)$, $i = 1, \dots, M$, and*
- (ii) *$x_{1t} = 1: t = 1, 2, \dots$ and $\lim_{T \rightarrow \infty} T^{-1}X'X = Q$, a positive definite matrix.*

Then,

$$\sqrt{T}[\xi_T^*(\theta_1) - \xi(\theta_1), \dots, \xi_T^*(\theta_M) - \xi(\theta_M)]$$

converges in distribution to an (MK)-variate Gaussian random vector with mean 0 and covariance matrix $\Omega(\theta_1, \dots, \theta_M; F) \otimes Q^{-1}$, where Ω is the covariance matrix of the corresponding M ordinary sample quantiles from random samples from distribution F .

PROOF: The proof for $M = 1$ is a trivial modification of the proof given in [6] for the special case $\beta^*(1/2)$. The case $M = 2$ is treated below explicitly, but the generalization to arbitrary M is obvious, albeit somewhat tedious. Consider the probability element,

$$(4.1) \quad g_T(\delta_1, \delta_2) d\delta_1 \cdot d\delta_2 = \text{pr} [\delta_i < \sqrt{T}(\xi_T^*(\theta_i) - \xi(\theta_i)) < \delta_i + d\delta_i, i = 1, 2].$$

We demonstrate that the joint density function $g_T(\delta_1, \delta_2)$ converges to a specified Gaussian density and Scheffé's theorem on convergence of densities completes the proof.

By Theorem 3.2,

$$(4.2) \quad \beta^*(\theta, u, X) = \beta^*(\theta, y, X) - \beta$$

where $u = y - X\beta$ is a vector of T independent realizations from F . So, by

Theorem 3.1 and the uniqueness of $\{\beta_T^*(\theta_1), \beta_T^*(\theta_2)\}$ there must exist index sets h_1, h_2 such that

$$(4.3) \quad \delta_i < \sqrt{T}(X(h_i)^{-1}u(h_i) - \xi(\theta_i)) < \delta_i + d\delta_i, \quad i = 1, 2,$$

and

$$(4.4) \quad (\theta_i - 1)\iota'_K < \sum_{t \in \bar{h}_i} [1/2 - 1/2 \operatorname{sgn}(u_t - x_t X(h_i)^{-1}u(h_i)) - \theta_i] x_t X(h_i)^{-1} < \theta_i \iota'_K, \quad i = 1, 2.$$

Since the first column of X is the unit vector, the events of (4.3) may be written as

$$(4.5) \quad u_t \in (\xi(\theta_i) + T^{-1/2}x_t \delta_i, \xi(\theta_i) + T^{-1/2}x_t(\delta_i + d\delta_i))$$

for $t \in h_i, i = 1, 2$. By hypothesis $\xi(\theta_1) < \xi(\theta_2)$ so (4.5) implies that for sufficiently large T , the solution index sets must be disjoint. Thus proceeding as in [6], we may write,

$$(4.6) \quad g_T(\delta_1, \delta_2) d\delta_1 \cdot d\delta_2 = \sum_{\substack{h_1 \in H \\ h_1 \cap h_2 = \emptyset}} \sum_{h_2 \in H} \operatorname{pr} [\delta_i < \sqrt{T}(X(h_i)^{-1}u(h_i) - \xi(\theta_i)) < \delta_i + d\delta_i, i = 1, 2] \\ \cdot \operatorname{pr} [(\theta_i - 1)\iota'_K < \sum_{t \in \bar{h}_i} z_{ti} < \theta_i \iota'_K, i = 1, 2]$$

where

$$z_{ti} = [1/2 - 1/2 \operatorname{sgn}(u_t - \xi(\theta_i) - T^{-1/2}x_t \delta_i) - \theta_i] x_t X(h_i)^{-1}.$$

But since $h_1 \cap h_2 = \emptyset$ and the u 's are i.i.d.,

$$(4.7) \quad \operatorname{pr} [\delta_i < \sqrt{T}(X(h_i)^{-1}u(h_i) - \xi(\theta_i)) < \delta_i + d\delta_i, i = 1, 2] \\ = T^{-K} |X(h_1)| |X(h_2)| \prod_{t \in h_1} f(\xi(\theta_1) + T^{-1/2}x_t \delta_1) \prod_{t \in h_2} f(\xi(\theta_2) \\ + T^{-1/2}x_t \delta_2) d\delta_1 \cdot d\delta_2.$$

Now note that the random variables,

$$(4.8) \quad z_{ti} = \begin{cases} -\theta_i x_t X(h_i)^{-1} \\ (1 - \theta_i) x_t X(h_i)^{-1} \end{cases} \quad \text{with probability} \quad \begin{cases} 1 - F(\xi(\theta_i) + T^{-1/2}x_t \delta_i), \\ F(\xi(\theta_i) + T^{-1/2}x_t \delta_i), \end{cases}$$

for $t \in \bar{h}_1 \cap \bar{h}_2$. And by (4.5),

$$z_{t1} = (1 - \theta_1) x_t X(h_1)^{-1}, \quad t \in h_2,$$

$$z_{t2} = -\theta_2 x_t X(h_2)^{-1}, \quad t \in h_1,$$

for sufficiently large T .

A bit of calculation reveals that the stabilized sums

$$Z_{Ti} = T^{-1/2} \sum_{t \in \bar{h}_i \cap \bar{h}_2} z_{ti}, \quad i = 1, 2,$$

converge to a $(2K)$ -variate Gaussian random vector with mean $(f(\xi(\theta_1))\delta'_1 QX'(h_1)^{-1}, f(\xi(\theta_2))\delta'_2 QX'(h_2)^{-1})$ and covariance matrix

$$\begin{bmatrix} \theta_1(1-\theta_1)X'(h_1)^{-1}QX(h_1)^{-1} & \theta_1(1-\theta_2)X'(h_1)^{-1}QX(h_2)^{-1} \\ \theta_1(1-\theta_2)X'(h_2)^{-1}QX(h_1)^{-1} & \theta_2(1-\theta_2)X'(h_2)^{-1}QX(h_2)^{-1} \end{bmatrix}.$$

This can be verified by expanding F around the population quantiles $\xi(\theta_i)$: $i = 1, 2$, noting that $T^{-1} \sum_{t \in \bar{h}_1 \cap \bar{h}_2} x'_t x_t \rightarrow Q$, and

$$T^{-1/2} \max_{k, t < T} |x_{kt}| = o(1);$$

see Malinvaud [30, 226–27]. It then follows that

$$\begin{aligned} (4.9) \quad T^K \text{pr} [-T^{-1/2}(\theta_i - 1)\iota'_K < Z_{Ti} < T^{-1/2}\theta_i \iota'_K, i = 1, 2] \\ = (2\pi)^{-K} |\Theta|^{-K/2} |X'(h_1)^{-1}QX(h_1)^{-1}|^{-1/2} |X'(h_2)^{-1}QX(h_2)^{-1}|^{-1/2} \\ \cdot \exp \{-\tfrac{1}{2}\delta'[\Omega^{-1} \otimes Q]\delta\} + o(1), \end{aligned}$$

where $\Omega = \Omega(\theta_1, \theta_2; F)$, $\delta = (\delta_1, \delta_2)'$, and

$$\Theta = \begin{bmatrix} \theta_1(1-\theta_1) & \theta_1(1-\theta_2) \\ \theta_1(1-\theta_2) & \theta_2(1-\theta_2) \end{bmatrix}.$$

The continuity of the density at $\xi(\theta_1)$ and $\xi(\theta_2)$ implies

$$(4.10) \quad \prod_{t \in \bar{h}_i} f(\xi(\theta_i) + T^{-1/2}x_t \delta_i) = f(\xi(\theta_i))^K + o(1), \quad i = 1, 2.$$

Substituting (4.10) and (4.9) into (4.6), we have

$$\begin{aligned} (4.11) \quad g_T(\delta) = \sum_{\substack{h_1 \in H \\ h_1 \cap h_2 = \emptyset}} \sum_{h_2 \in H} T^{-2K} |X(h_1)|^2 |X(h_2)|^2 f(\xi(\theta_1))^K f(\xi(\theta_2))^K |\Theta|^{-K/2} |Q|^{-1} \\ \cdot \exp \{-\tfrac{1}{2}\delta'[\Omega^{-1} \otimes Q]\delta\} + o(1) \sum_{h_1} \sum_{h_2} T^{-2K} |X(h_1)|^2 |X(h_2)|^2. \end{aligned}$$

But,

$$(4.12) \quad T^{-2K} \sum \sum |X(h_1)|^2 |X(h_2)|^2 = T^{-2K} |X'X|^2 = |T^{-1}X'X|^2$$

converges to $|Q|^2$; see Rao [34, p. 32]. So simplifying (4.11), we have

$$g_T(\delta) \rightarrow (2\pi)^{-K} |\Omega^{-1} \otimes Q|^{1/2} \exp \{-\tfrac{1}{2}\delta'[\Omega^{-1} \otimes Q]\delta\}.$$

And finally, Scheffe's [37] theorem on convergence of densities yields the desired conclusion.

REMARK: An extremely important special case is the regression median, or least absolute error estimator, $\beta^*(1/2)$. Without loss of generality β may be located so that $F(0) = 1/2$, so $\xi(1/2) = 0$. Then the asymptotic distribution of the random variable $\sqrt{T}(\beta^*(1/2) - \beta)$ is K -variate Gaussian with mean zero and covariance matrix $[2f(0)]^{-2}Q^{-1}$. See our detailed treatment of this special case in [6]. Thus, *the regression median is seen to be more efficient than the least squares estimator in the linear model for any distribution for which the median is more efficient than the mean in the location model*. While such a result has been the subject of conjecture among a number of authors we believe ours to be its first formal statement.¹⁷

The remarkable parallelism between the asymptotic behavior of ordinary sample quantiles in the location model and regression quantiles in the linear model suggests a straightforward extension of the large sample theory of so-called systematic statistics from the location model to the linear model. This is made explicit in the following result.

THEOREM 4.3: *Let $\pi(\theta) = (\pi(\theta_1), \dots, \pi(\theta_M))'$ be a discrete, symmetric probability measure on $(0, 1)$ concentrating mass on $\{\theta_i: i = 1, \dots, M | 0 < \theta_1 < \theta_2 < \dots < \theta_M < 1\}$. Suppose $F(0) = 1/2$, so $\xi(1/2) \equiv F^{-1}(1/2) = 0$, and conditions (i) and (ii) of Theorem 4.2 hold. Then*

$$\tilde{\beta}_T(\pi) = \sum \pi(\theta)\beta_T^*(\theta)$$

is invariant to location, scale and reparameterization of design in the sense of Theorem 3.2, and $\sqrt{T}(\tilde{\beta}_T(\pi) - \beta)$ converges in distribution to a (K) -variate Gaussian random vector with mean zero and covariance matrix $\pi' \Omega \pi \cdot Q^{-1}$.

PROOF: Immediate from Theorems 3.2 and 4.2.

REMARK: Obvious examples of weight functions $\pi(\theta)$, in addition to the regression median example ($\pi(1/2) = 1$) already discussed, are the Gastwirth: $(\pi(1/3), \pi(1/2), \pi(2/3)) = (.3, .4, .3)$ and the trimean: $(\pi(1/4), \pi(1/2), \pi(3/4)) = (1/4, 1/2, 1/4)$. When F is non-Gaussian, weight functions $\pi(\theta)$ are readily found which make $\pi' \Omega \pi$ smaller than the variance of F , i.e., yielding estimators $\tilde{\beta}(\pi)$ having superior asymptotic efficiency to the least squares estimator. Furthermore, as we have noted above, many estimators of the form $\tilde{\beta}(\pi)$ like Gastwirth's and the trimean, have high efficiency over a large class of distributions. The efficiency of the mean and its least squares analogue, $\hat{\beta}$, in the

¹⁷ In finite samples the form of the design matrix obviously has some impact on the sampling distribution of the estimates, and is likely therefore to affect the sample size required to achieve a given degree of convergence to the asymptotic distribution. Thus there is no contradiction between our general result and the finding of Rosenberg and Carlson [36] that different designs have somewhat different sampling distributions in small sample Monte-Carlo experiments. Rosenberg and Carlson's results do seem to suggest the kurtosis of the explanatory variables produces a slower convergence. These problems of robustness of design are only now beginning to receive some much deserved attention. See, e.g., the pioneering work of Huber [27] on minimax designs.

linear model, on the contrary, deteriorates rapidly as the error distribution tends away from the Gaussian toward longer tailed distributions.

The contrast between the ordinary least squares estimator and linear combination of regression quantiles, $\tilde{\beta}(\pi)$ -type, estimators for the linear model is highlighted by writing the former in the partitioning notation of Section 3 as

$$\hat{\beta} = \sum w(h) \hat{\beta}(h),$$

where $w(h) = |X(h)|^2 / \sum |X(h)|^2$, summations are over $h \in \mathcal{H}$, and $\hat{\beta}(h)$ satisfies the normal equations,

$$X(h) \hat{\beta}(h) = y(h).$$

This remarkable formulation of the least squares estimate as a weighted average of all possible coefficient vectors defined by subsamples of K observations was apparently first noted by M. Subrahmanyam [40]. Least squares places positive weight on *all* $\hat{\beta}(h)$ with nonzero design determinant; the $\tilde{\beta}(\pi)$ estimators place positive weight on only a few select $\hat{\beta}(h)$ vectors. The simple connection between the sample mean and the sample quantiles in the location model thus generalizes in a somewhat more sophisticated form to the connection between the least squares estimator $\hat{\beta}$ and the class of $\tilde{\beta}(\pi)$ estimators for the linear model.

In a very stimulating critique of least squares estimation procedures, Tukey [42] suggests that efforts should be made to find estimators which modify the least squares method, reducing its notorious sensitivity to outlying observations, but preserving its essentially good qualities. In this spirit we discuss a simple modification of least squares which employs regression quantiles.

An obvious analogue to the trimmed mean of the location model is suggested by regression quantiles. Let α be the desired trimming proportion. Calculate $\beta^*(\alpha)$ and $\beta^*(1-\alpha)$ for the sample. By Theorem 3.4 there will be approximately $T\alpha$ observations below the former and above the latter. These observations are set aside, and the remaining observations are subjected to a least squares fit. Since the β^* vectors are consistent estimators of the α and $(1-\alpha)$ population quantile hyperplanes, least squares on the "censored" sample is (asymptotically) like sampling from the distribution F truncated at $F^{-1}(\alpha)$ and $F^{-1}(1-\alpha)$. This estimator, say $\hat{\beta}_\alpha$, is location and scale invariant, and will be asymptotically K -variate Gaussian with covariance matrix $\sigma^2(\alpha, F)Q^{-1}$ where $\sigma^2(\alpha, F)$ denotes the asymptotic variance of the corresponding α -trimmed mean from a population with distribution F . This class offers promising robustness properties and should behave similarly to Huber's (M) estimates. Computation of these trimmed least squares estimates merely requires the solutions of two simple linear programming problems in addition to the usual least squares computation. The choice of the trimming proportion α will depend, obviously, on one's confidence in the quality of available data. With data of very dubious reliability one may want to use the ultimate trimmed least squares estimator with $\alpha = 1/2$, the regression median or LAE estimator. For somewhat better data, a trimming proportion of five to ten per cent is probably preferable.

5. CONCLUSION

We argue that the conventional least squares estimator may be seriously deficient in linear models with non-Gaussian errors. In the absence of a well-specified prior on the set of plausible distribution functions it is useful, following Huber [25] and others, to view the problem of estimation in terms of insurance. It seems reasonable to pay a small premium in the form of sacrificed efficiency "at the Gaussian distribution" (if that is the hypothesized parametric model), in order to achieve a substantial improvement over least squares in the event of a non-Gaussian situation.

We introduce a new class of statistics for the linear model which we have called "regression quantiles" since they appear to have analogous properties to the ordinary sample quantiles of the location model. Natural generalizations based on regression quantiles of linear combinations of sample quantiles and trimmed means which appear to have promising robustness properties are then proposed for the general linear model.

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APPENDIX

COMPUTATIONAL ASPECTS OF REGRESSION QUANTILES¹⁸

The regression quantile minimization problem posed above in Section 3 is equivalent to the linear program

$$(P) \quad \min [\theta \iota' r^+ + (1 - \theta) \iota' r^-]$$

subject to

$$\begin{aligned} y &= Xb + r^+ - r^-, \\ (b, r^+, r^-) &\in \mathbb{R}^K \times \mathbb{R}_+^{2T}, \end{aligned}$$

where $\iota' = (1, 1, \dots, 1)$, a T vector of ones. The dual to (P) is the bounded variables problem,

$$(D) \quad \max [y'd]$$

subject to

$$\begin{aligned} X'd &= 0, \\ d &\in [\theta - 1, \theta]^T, \end{aligned}$$

where $[\theta - 1, \theta]^T$ denotes the T -fold Cartesian product of the closed interval $[\theta - 1, \theta]$. It proves convenient to make one more minor adjustment. Translate the dual variables setting $\Delta = d + 1 - \theta$, so we have

$$(D') \quad \max [y'\Delta]$$

¹⁸ For further details on the important special case, $\theta = 1/2$, see [1, 43].

subject to

$$\begin{aligned} X'\Delta &= (1 - \theta)X'\epsilon, \\ \Delta &\in [0, 1]^T. \end{aligned}$$

This translated dual formulation proves to be extremely convenient for computational purposes. Standard linear programming algorithms provide quite efficient algorithms for such bounded variables problems. So-called "parametric variation of the right-hand-side" of (D') permits economical solution for many, or indeed all, $\theta \in [0, 1]$.

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Quantile regression: A short story on how and why

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Abstract: Quantile regression quantifies the association of explanatory variables with a conditional quantile of a dependent variable without assuming any specific conditional distribution. It hence models the quantiles, instead of the mean as done in standard regression. In cases where either the requirements for mean regression, such as homoscedasticity, are violated or interest lies in the outer regions of the conditional distribution, quantile regression can explain dependencies more accurately than classical methods. However, many quantile regression papers are rather theoretical so the method has still not become a standard tool in applications. In this article, we explain quantile regression from an applied perspective. In particular, we illustrate the concept, advantages and disadvantages of quantile regression using two datasets as examples.

Key words: quantile regression, tutorial article, additive regression models, gradient boosting

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1 Introduction

Mathematically speaking, the probable [...] and the improbable [...] are not different in kind, but only in frequency, whereby the more frequent appears a priori more probable. But the occasional occurrence of the improbable does not imply the intervention of a higher power, something in the nature of a miracle, as the layman is so ready to assume. The term probability includes improbability at the extreme limits of probability, and when the improbable does occur this is no cause for surprise, bewilderment or mystification.

(Frisch, 1959)

This is a quote from Max Frisch's most famous book *Homo Faber* in which the main character meets his daughter, whose existence he did not know of, and falls in love with her. This is only one of the many coincidences in this book, or—as he puts it—one of the improbable events. Another one is a plane crash, which leads him to the earlier quoted reasoning. This example is not very close to our everyday life experience. As

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an introduction to the topic of this tutorial article, we hence use a dataset on the body mass index (BMI) of Dutch males aged 0 to 21 (from now on referred to as the ‘Dutch boys’ dataset). The data will be described in detail in Section 2. The average BMI of the 7 294 observations is 18.027. Thus, we can say ‘the expected BMI for a dutch male between 0 and 21 years is’ 18.027. However, reducing an experiment to its expectation is exactly what the main character of our novel criticizes the ‘layman’ for, who is surprised by the occurrence of events at the ‘the extreme limits of probability’. Furthermore, those extreme limits are highly relevant for the BMI data since obesity and (in case of the Dutch population maybe less relevant) underweight are more relevant than the simple question for the average weight. But what are those ‘limits of probability’ and how can we capture them? A histogram of the data, together with a fitted Gaussian distribution is displayed in the left panel of Figure 1. Distributions can be described quite accurately by their moments, that is, expectation, variance, skewness and so on, as well as by a list of quantiles.

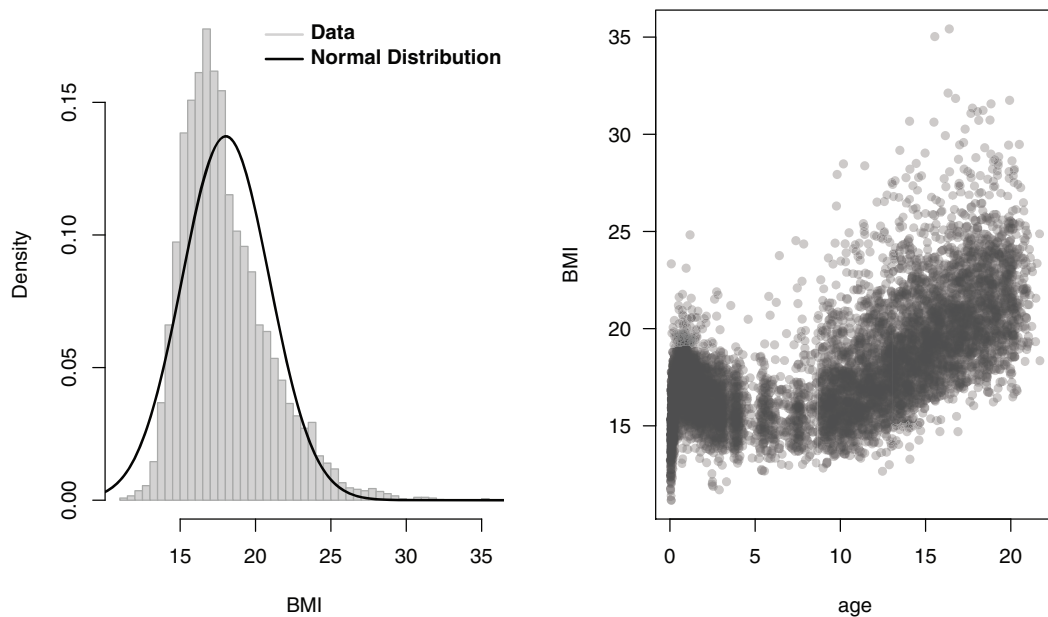


Figure 1 Histogram of the BMI from the Dutch boys dataset with the Gaussian distribution with mean and standard deviation taken from the dataset

In the case of the Gaussian distribution, just as for most distributions from a parametric family, those statistics are known and can be calculated for arbitrary

parameters. As we can see in the left panel of Figure 1, however, the shape of the fitted Gaussian distribution is quite different from that of the observed distribution of the data, as the histogram is quite asymmetric in comparison to the distribution curve. When one is interested in the extreme values of the dataset (e.g., the 95% quantile), we would hence refrain from using the quantiles calculated via the distribution and simply use the quantile of the data itself: while the 95% quantile of the fitted normal distribution is 22.809, the 95% quantile of the data is 23.612. This is a relevant difference; the approximation with the Gaussian distribution would substantially misinform us. Researchers, however, hardly ever only deal with the simple description of the univariate dataset, but are more interested in the influence of covariates on a dependent variable or the prediction based on independent variables. In our case, we could imagine wanting to predict the BMI depending on the age of an individual. The plot in the right panel of Figure 1 displays the scatterplot of age and BMI. The asymmetry already visible in the univariate representation also shows in the scatterplot. When doing classical regression on this relation however, we would assume exactly the same thing as when plainly approximating our histogram with a Gaussian distribution: that the BMI conditional on the age is normally distributed and this assumption hence underlies all statistics we derive from this (such as the conditional 95% quantile). The solution to this problem is to use quantile regression, that is, calculating the impact of covariates on quantiles directly, rather than assuming an underlying conditional distribution. This tutorial article explains what quantile regression is and how it can be calculated.

The remainder of the article is structured in the following way: the second section deals with univariate quantile regression model illustrated by the aforementioned dataset on the BMI of Dutch boys. The dataset will be analysed with both standard regression method as well as with quantile regression and the differences will be discussed. The third section will present a second example dataset, which is then used to show an additive quantile regression model, containing different types of covariates. In the fourth section different estimation methods and related models will be introduced. The last section consists of a short summary and a guideline of when to use quantile regression.

The supplementary material to this article includes the commented code for both examples. They are estimated with component-wise gradient boosting (Hofner et al., 2017); for more detailed explanation for this method please see the tutorial on boosting (Mayr and Hofner, 2018).

2 Basic quantile regression

2.1 Data example I: The BMI in the Netherlands

As a first illustrative example, we present the Fourth Dutch Growth Study which was already mentioned in the introduction. The dataset originates from a cross-sectional study that measures growth and development of the Dutch population between 0 and 21 years. We use a subset of the data available in the R package `gamlss`

(Stasinopoulos et al., 2017), which includes 7 294 observations of male Dutch children. In the following, we will refer to the dataset as ‘Dutch boys’ dataset.

The variable of interest in this example will be the BMI of the individuals with a special focus on those who are overweight. Recall Figure 1, where the fitted Gaussian distribution yields $P(\text{BMI} > 30) = 1 - F(\text{BMI} < 30) = 1.91 \cdot 10^{-5}$, while the empirical probability (i.e., the proportion of participants with $\text{BMI} > 30$) is 0.0021, which is two orders of magnitude larger. This is something that already shows in the comparison of the histogram to the theoretical distribution: the data deviates substantially from a Gaussian distribution. But what if we ask questions such as: Which BMI values should we consider to be comparatively high? Where on the scale are the 5% of children with the highest BMI? Which value does y have to have such that

$$P(\text{BMI} < y) = 95\%$$

holds? The correct answer to those questions is the 95% quantile of the distribution, which can be defined in the following way: ‘the τ quantile is the value y , for which 100 τ % of the values are smaller or equal than this X ’ or (in a formula):

$$Q_\tau(y) = F_y^{-1}(\tau). \quad (2.1)$$

The 95% quantile of the fitted normal distribution is 22.809, the 95% quantile of the data is 23.612. Hence, 5% have a higher BMI, and the rest has a lower BMI. Again theoretical and empirical result differs a lot, the distribution can thus not be considered to be Gaussian.

When only analysing the BMI, we ignore the fact that its conditional distribution varies with age, as can be seen in Figure 1 on the right side. The questions asked should not be restricted to the simple quantile (2.1) but be a quantile conditioned on age:

$$Q_\tau(y|x) = F_{y|x}^{-1}(\tau|x) = y_\tau(x), \quad (2.2)$$

where x is age. The most simple explanation for this formula is that for each τ it determines a function $y_\tau(x)$ bisecting the data such that, for any age x , 100 τ % of the data lie below the value of the function at that age. The function defined in this way splits the data into a τ and a $1 - \tau$ proportion, which is based on the covariate age. This model was first established by Koenker and Bassett (1978) by introducing a linear quantile regression and explaining a first algorithm that is able to calculate the parameters desired to choose a β such that

$$Q_\tau(y|X) = X\beta_\tau,$$

with X being the covariate matrix just as in usual mean regression. In order to cut off 100 τ % proportions sensibly in the case of the plot in the right panel of Figure 1, the quantile regressions cannot simply consist of linear functions, since the relation between age and BMI does not seem to be a linear one. In the following, we will use

penalized splines (P-splines) (Eilers and Marx, 1996) in order to model the different regression curves. P-splines are a special way of modelling non-linear effects. For a detailed explanation see Fahrmeir et al. (2013).

2.2 Modelling the BMI of Dutch boys

In the following, both mean regression and quantile regression models will be applied to the presented dataset.

2.2.1 Mean regression

The conventional non-linear regression model estimating the association of age with the expected value of BMI was calculated with R-package *mgcv* (Wood, 2014). As can be seen in Figure 2, the resulting curve lies in the centre of the data, but is unaffected by the asymmetry of the data, which is especially pronounced at higher ages.

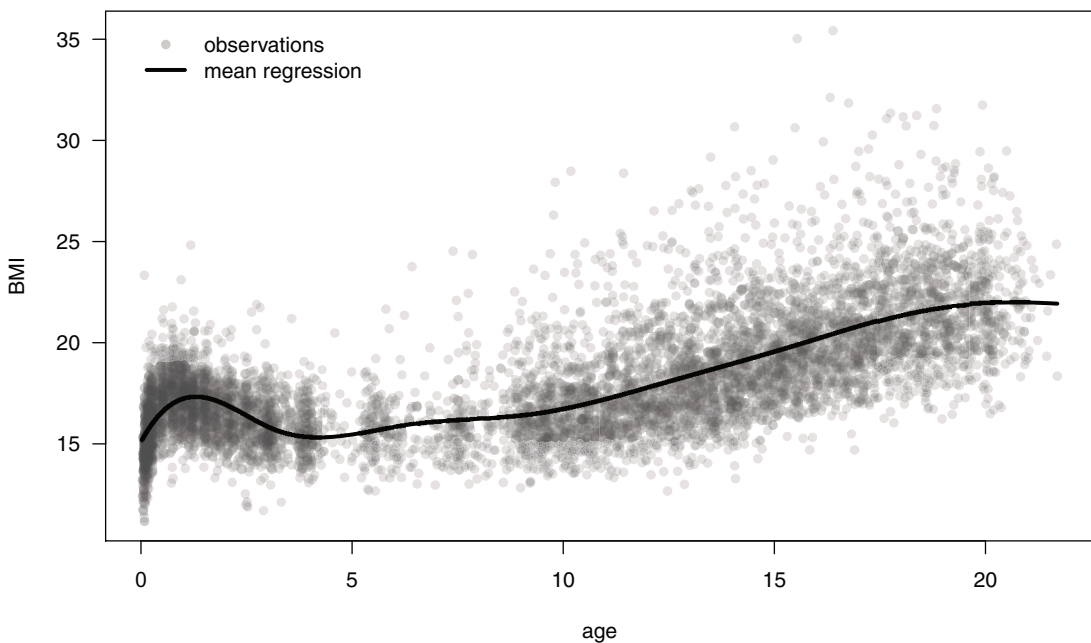


Figure 2 Scatterplot of age and BMI of the dataset Dutch boys with the regression line from a non-linear Gaussian regression model

The assumptions for conventional mean regression models which are violated quite obviously in this dataset are homoscedasticity and the symmetry of the Gaussian distribution. In addition to the mean curve, the quantiles which are

implied by the assumption of a Gaussian distribution are displayed for $\tau = (0.01, 0.1, 0.2, 0.3, 0.4, 0.6, 0.7, 0.8, 0.9, 0.99)$. These are calculated based on the variance obtained, when assuming a Gaussian distribution for the errors. These quantile curves do not fit the data very well for low and high BMI, which makes it even clearer that the assumption of parallel quantiles implied by the constant variance assumed in conventional regression is not met at all in the data, yet it is assumed in the requirements of the mean regression model. The BMI of the boys varies more the older they get. This increase in variation is not symmetrical. While the data is only slightly less dense for low BMIs, it is quite sparse in higher BMIs. It is unfavourable that exactly the part of the distribution which is of medical interest (i.e., potentially overweight children) is captured so badly.

2.2.2 *Quantile regression*

When modelling the quantiles independently of distributional assumptions yet conditional on the data (i.e., $Q(Y|X)$), we get the model as displayed in the right panel of Figure 3. The quantile-specific regression curves show differences to the ones extracted from the aforementioned Gaussian model in both aspects. The asymmetry of the distribution is captured by the approach. If the distribution was symmetrical, the distance between the median regression and the curve for $\tau = 0.6$ should be the same as the difference between the median regression and the curve for $\tau = 0.4$. The distance between $\tau = 0.4$ and $\tau = 0.3$ should be the same as the difference between $\tau = 0.6$ and $\tau = 0.7$, etc. This assumption does not hold in any of the quantile curves. The difference which is most obvious can be seen in the distance between the first and second (i.e., the $\tau = 0.01$ and the $\tau = 0.1$) quantile and the distance between the second to last and the last (i.e., the $\tau = 0.9$ and the $\tau = 0.99$) quantile. This reflects the positive skewness of the data which was already visible in Figure 1 in the left panel. The second central difference to the model shown earlier is the capturing of the heteroscedasticity. Again, the effect is most pronounced in the extremal quantiles: the increasing difference between the $\tau = 0.01$ quantile and the $\tau = 0.99$ quantile shows drastically how the variance in the BMI increases with increasing age. The difference in behaviour of the curves ultimately shows the importance of the approach: deviations from a conditional normal distribution could not have been captured without it. A second visualization of the fact that the conditional distribution is skewed is Figure 4, in which we can see that the mean is slightly above the median.

A third feature of quantile regression is the robustness against outliers. As can be seen in Figure 4, the median regression simply divides the data into two 50% parts. This figure also shows that the values scattered further away from the dense central cloud (e.g., those close to 35) only have an influence on the 99% quantile regression curve and not the more central curves. This, however, is not a problem since those are the data points that should be captured and described when looking at the 1% of the population with the highest BMI. The fact that the mean regression is above the 50% quantile regression curve is due to the long tail of the conditional distribution of the BMI.

3 Additive quantile regression

3.1 Data example II: Stunting in India

For the second data example, a dataset called `india` from the R package `gamboostLSS` (Hofner et al., 2017) was used (please note that the original dataset is publicly available on <http://www.measuredhs.com> after registration). It includes a number of outcomes from an Indian study on malnutrition in children up to the age of 3 which was conducted in the years 1998 and 1999. Since the process of growth is inhibited by malnutrition, stunting itself can be used as indicator for nutritional status of children. The outcome of interest in this case is hence stunted growth represented by a z-score measuring growth ranging from -6 to 6 , where negative values mean that a child is too short for its age, which is an indicator of prolonged malnutrition. Children with values below -2 are considered stunted (height-for-age). Since India has a severe problem with malnutrition among children, the interest in this dataset is very different from the the Dutch dataset. While the focus in the previous section was on analysing the higher quantiles due to the interest in risk of overweight, in this case,

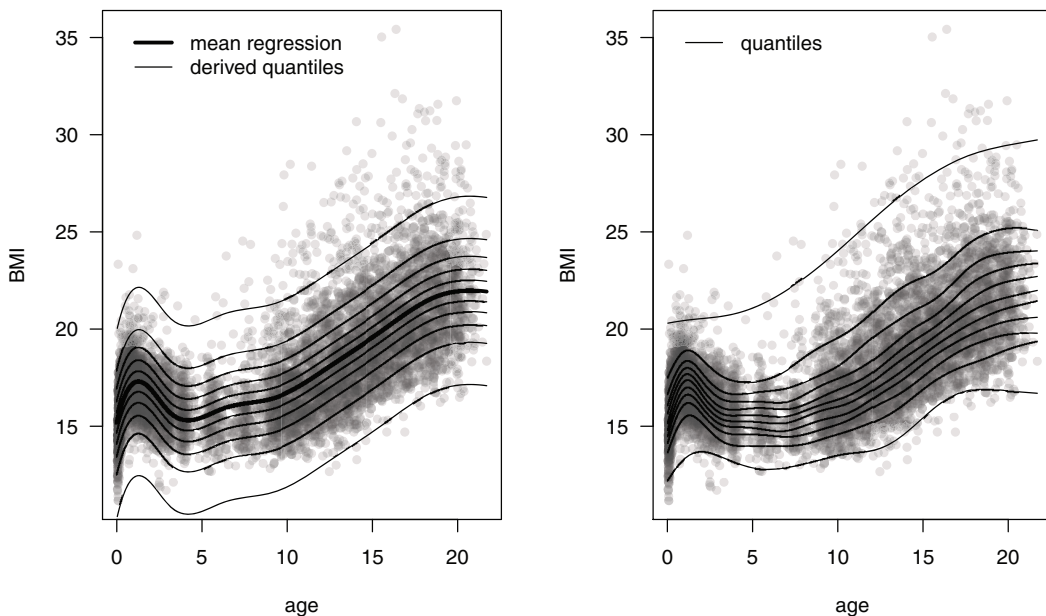


Figure 3 Scatterplot of age and BMI of the dataset Dutch boys. Left hand side: the bold line is the regression line from a non-linear Gaussian regression model, the other lines depict the $\tau = (0.01, 0.1, 0.2, 0.3, 0.4, 0.6, 0.7, 0.8, 0.9, 0.99)$ quantiles of the mean regression model. Right hand side: the lines depict the $\tau = (0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.99)$ quantile regression models

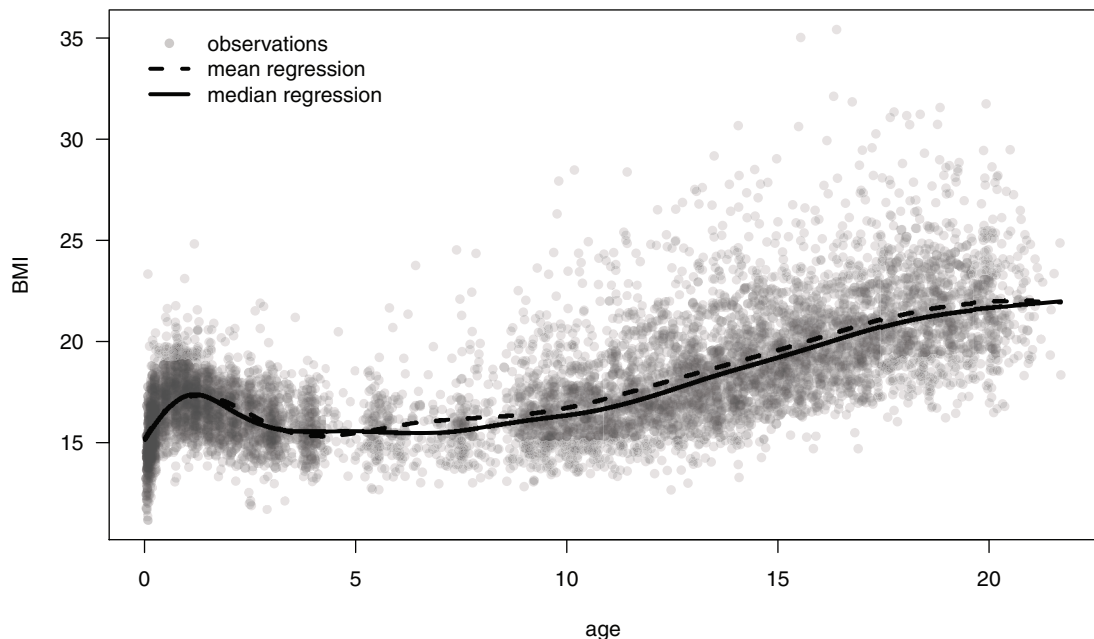


Figure 4 Scatterplot of age and BMI of the dataset Dutch boys. The solid line displays the median regression and the dashed line displays the mean regression

we are mainly interested in analysing the lower part of the distribution. The dataset contains the ‘z-score’ of 4 000 children, and the covariate ‘age’ in months, the ‘BMI of the mother’ and the ‘district’ the child was living in at the time of the study. The model described in the following section will measure the impact of the covariates on the z-score. For a further analysis of this dataset with a more content-related interpretation, see for example Fenske et al. (2013). The same data example is used in the tutorial article on boosting (Mayr and Hofner, 2018).

3.2 Modelling the Z-score of Indian children

The model presented in the previous section (i.e., a model with one continuous explanatory variable) is a comparatively simple explanatory model. Quantile regression methods were extended, however, in many different ways towards more complex model classes, just like mean regression, such as for models with measurement errors by Wei and Carroll (2009), time series by Kley et al. (2016), or additive, partly non-linear models by Lee et al. (2010). For an overview over the progress that has been made, see the paper of Koenker (2017); for the range of types of effects that can be included in structured additive quantile regression, see Kneib (2013). In the following, we present an additive model for the dataset on stunting in

India described earlier. The model includes three covariates, being modelled in two different ways. While the influence of the age of the child—just as for the Dutch boys dataset—and the BMI of the mother will be modelled by P-splines (Eilers and Marx, 1996), the geographical differences will be modelled by a Markov random field. This method allows all districts to have their own effect on the dependent variable (or in our case the quantile of interest of the dependent variable), but it does not allow neighbouring districts to vary too much from each other. This leads to a similar type of penalization as in P-splines and thus also to a smooth surface (Fahrmeir et al., 2004). The model formula hence looks like this:

$$Q_{\tau}(y_{\text{stunting}}|X) = \beta_0 + f_{\text{tnl}}(x_{\text{age}}) + f_{\text{tnl}}(x_{\text{BMI}}) + f_{\text{r spat}}(x_{\text{district}}),$$

where β_0 is an intercept and the f stands for non-linear (nl) and spatial (spat) functions. Due to the predictor being a sum of different functions, this type of model is called ‘additive model’. There are many different ways to estimate the effects of those kinds of models, three of which will be listed in the next section. For this example, the component-wise gradient ‘boosting’ approach (Fenske et al., 2011) with the following call was used:

```
qr_boost <- gamboost(stunting ~
  age + BMI +
  bmrfl(district, "markov", bnd=india.bnd, center=F),
  family = QuantReg(tau = 0.3),
  data = india)
```

The call `gamboost` combined with the specification `family = QuantReg(tau = 0.3)` produces an additive quantile regression model for the $\tau = 0.3$ quantile. The algorithm is programmed such that it recognizes age and BMI to be continuous variables and thus automatically assigns them smooth non-linear effects. The inner function `bmrfl(district, "markov", bnd=india.bnd, center=F)` uses the regions in the variable `districts` which are linked to an additional data object `india.bnd`. This object is a so-called ‘boundary file’ which includes information on the districts and their boundaries. The “markov” tells the function to use the aforementioned Markov properties (i.e., values $f_{\text{r spat}}(x_{\text{district}})$ from neighbouring districts to be dependent in certain ways). The boosting algorithm can, besides estimating the effects, also carry out variable selection. This feature will not be described here in detail, but we note that it can be useful when dealing with a large number of possibly influential variables and being interested in variable selection in general. The models obtained by this algorithm have the advantage of a very good interpretability.

A set of models with different quantile levels $\tau = (0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.99)$ was calculated. The estimated effects for these models are illustrated by Figures 5–7. In Figure 5, the effect of age on the different quantiles

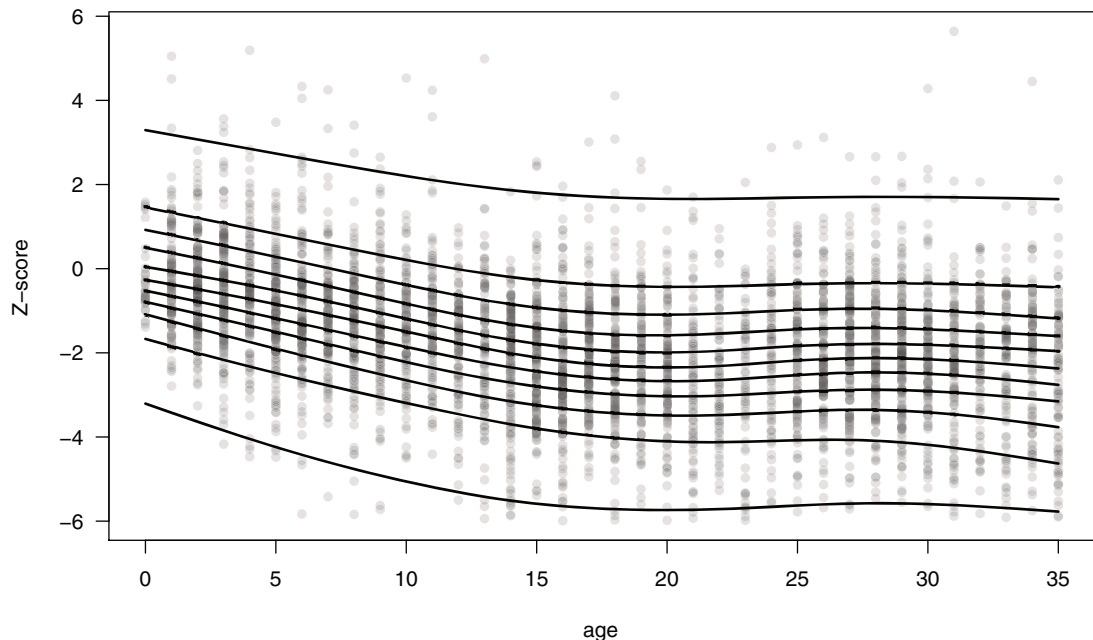


Figure 5 Scatterplot of age and z-score of the dataset on stunting in India. The lines depict the $\tau = (0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.99)$ quantile regression models adjusted for the rest of the data

is displayed. In order to be able to plot the non-linear effects into one graph, each quantile is shifted to the level of the predicted values for this quantile.

To this end, we use a centred version of the effect and then add the mean of the conditional quantile (i.e., the average over the prediction for all individuals) of interest. The different shape of the curves for different quantiles can be seen, just like in Section 2. Especially, the steeper decline for the lowest quantile ($\tau = 0.01$) in the early months attracts the attention. This indicates that malnutrition worsens with increasing age in the lower part of the distribution, that is, for children who are already in a more severe state. Figure 6 shows how in certain data situations, problems can occur. In this case, the effect of the BMI of the mother on the z-score is displayed. The estimation of the curves is increasing nearly linearly for lower BMI values, in most quantiles. In high BMI values, the central quantile, curves are flattening out. In the highest quantile, the impact of the BMI of the mother is rising until a BMI of 25 and then falling again. Estimations for higher levels of BMI of the mother are less accurate, since there are fewer observations.

The problem can be illustrated by a simple example: there are 7 women that have a BMI higher than 35. Thus, between 35 and 40, all 11 functions are estimated based on those 7 data points, which is very little information. We hence try to construct functions, which split 7 data points into 12 groups, without even informing one

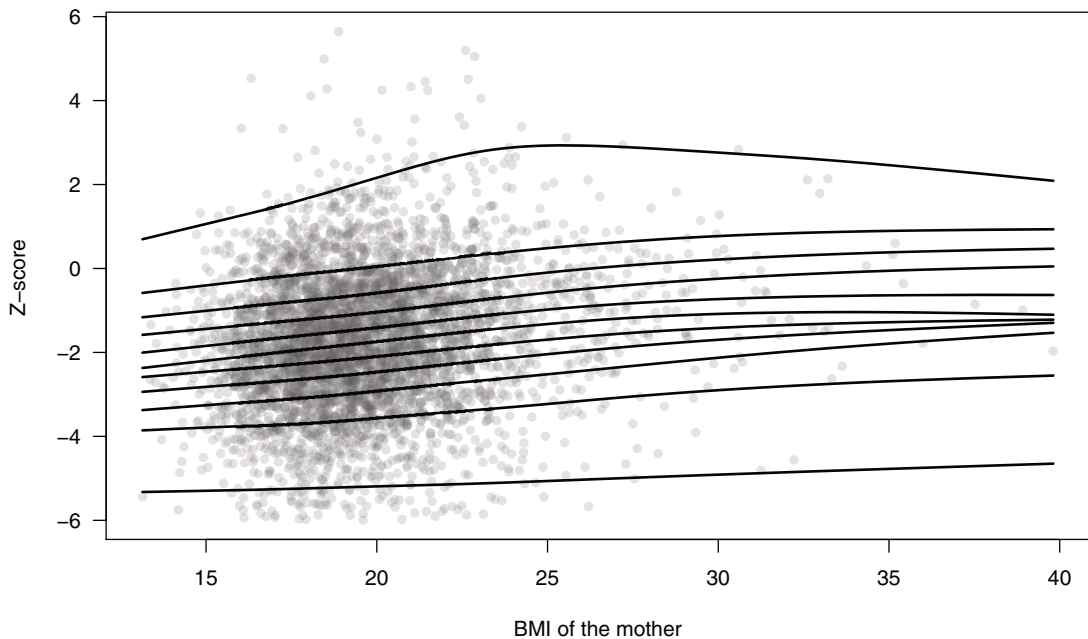


Figure 6 Scatterplot of the BMI of the mother and z-score of the dataset on stunting in India. The lines depict the $\tau = (0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.99)$ quantile regression models adjusted for the rest of the data

function about the position of the rest. When estimating the models independently for the different quantiles (as we did here), this can even lead to the quantiles to cross. For references to methods avoiding crossing see Section 4. Figure 7 displays the spatial effects for the quantiles $\tau = 0.2$, $\tau = 0.5$ and $\tau = 0.8$. In all three plots, the estimations are smooth in the sense that neighbouring districts do not differ very much in terms of effects. Between quantiles, however, a few differences are present. As an exemplary region Bihar was chosen. This region has a small frame around it in the upper row of the figure and is displayed in more detail in the lower row of the figure. While a negative trend can be spotted in the 0.2 and 0.5 quantiles, this effect is not visible at all in the 0.8 quantile.

4 Implementations and related models

In this article a gradient boosting approach was used to produce the results for the illustrative example. This could however have been done with many different approaches, which will be explained in the following.

Quantile regression has many advantages, but a major disadvantage is that parameters are harder to estimate than in Gaussian or generalized regression.

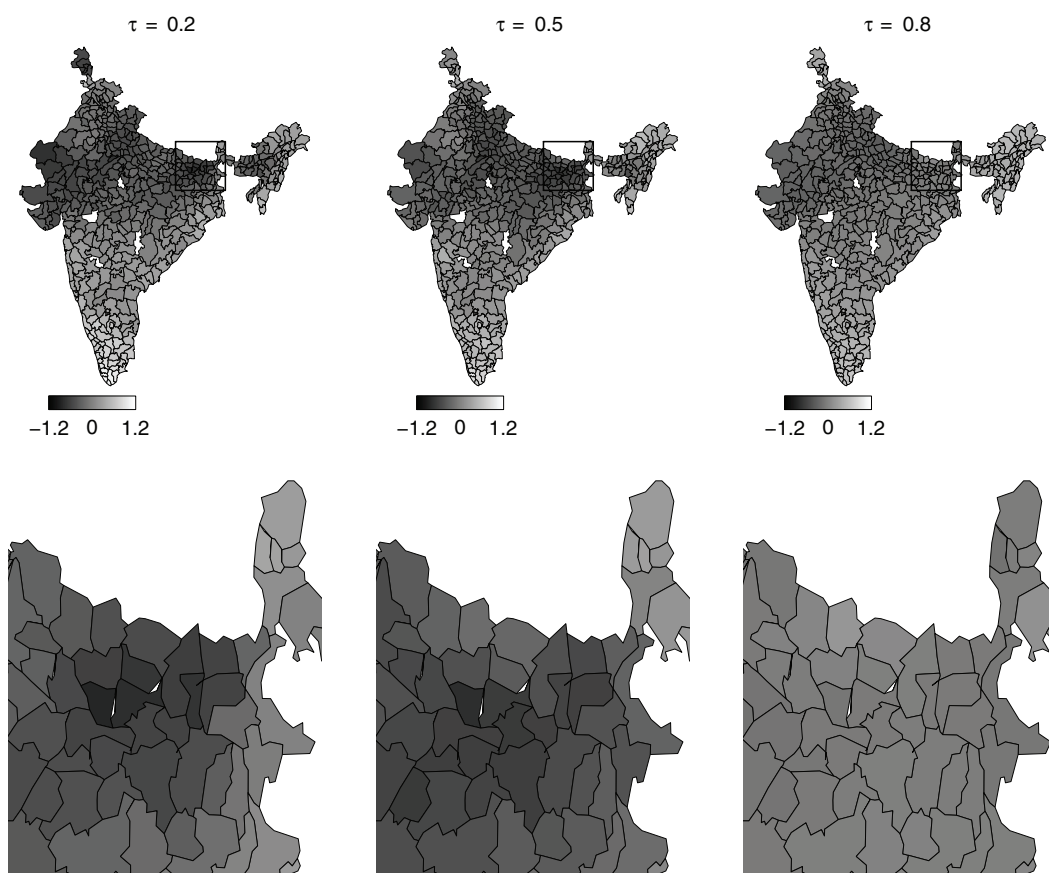


Figure 7 Effect of the district on the $\tau = 0.2$, the $\tau = 0.5$ and the $\tau = 0.8$ quantiles. The framed part contains the district of Bihar, which is further displayed in more detail

Inference on them can get complicated because the estimators for coefficients are not available in closed form. There are many ways to estimate quantile regression parameters; this will be sketched out in the following three ways. The first and original way is a linear optimization algorithm and was brought forward by Koenker and Bassett (1978) together with the original proposal. This approach is implemented in the R-package `quantreg` (Koenker, 2016), which includes linear as well as non-linear effects. Confidence intervals for the latter are based on piece-wise linear approximations and when using too many effects, the algorithm shows instabilities (Kneib, 2013). Model diagnostics rely on the construction of a statistic related to the R^2 in the mean regression case. The statistic is called R^1 and uses the minimizers of the check function for the models which have to be compared (Koenker and Machado, 1999). A second option is using boosting algorithms (Fenske et al., 2011) to

estimate sophisticated quantile regression models, which can handle a whole range of different effects, such as linear, non-linear and spatial effects; however, it lacks of a straight forward implementation of p-values and confidence intervals of the estimated regression parameters. This algorithm is implemented in the R package *mboost* (Hothorn et al., 2016). To find the optimal quantile regression model, however, appropriate tuning parameters like the number of iterations the algorithm has to perform have to be chosen. In the web-appendix, the example shown in this article is presented with comments on how to deal with those requirements. Boosting algorithms do not naturally produce p-values, but they can be generated post hoc. The gradient boosting algorithm is used in this article because the functions in the R package (Hothorn et al., 2016) are straightforward to apply. A third option is to use Bayesian inference (see e.g., Yu and Moyeed, 2001). An implementation of a structured additive model can be built in the stand-alone MCMC tool BayesX (Belitz et al., 2013). Due to the Bayesian structure, the algorithm delivers distributions of the estimated parameters. Uncertainty can hence be captured approximately, but variances of parameters are usually underestimated (Waldmann et al., 2013). Model diagnostics in Bayesian quantile regression do in general rely on the auxiliary distribution and not a real distributional assumption. Comparison between models is hence possible yet the comparison with models generated by other approaches is not possible. The Bayesian model involves a well-calibrated specification of the prior distributions and hyper parameters for optimal modelling. In most cases, they should be chosen as little informative as possible. For more information on how to chose appropriate priors and hyperprior parameters see, for example, Berger (2006). One major drawback of quantile regression is that most estimation techniques can lead to quantile crossing. This problem is mainly caused by the fact that those techniques estimate the models for all quantiles independently. There are, however, methods which prevent this problem, like for example the so-called quantile sheets Schnabel and Eilers (2013).

Another form of regression that exceeds the form of simply modelling the mean but does not have the issue of crossing quantiles is the generalized additive model for location, shape and scale (GAMLSS Rigby and Stasinopoulos, 2005). For a tutorial paper on this topic see Stasinopoulos et al. (2018). In this type of regression, all parameters of the assumed distribution can be modelled by covariates. The model is estimated as a whole, hence problems similar to the quantile crossing issue cannot occur in this case. This method is in some cases more stable than quantile regression (Kneib, 2013) and delivers similar results, as long as the assumed distribution is close enough to the real data distribution. This can be checked either graphically or by cross-validation. The Bayesian approach called distributional regression works analogously to this approach (Umlauf and Kneib, 2018). A related idea which has been explored less is the so-called Bayesian density regression (Dunson et al., 2007). In this case, the distribution of the dependent variable does not have to be specified beforehand but drawn in a Dirichlet process mixture scheme. A further model form closely related to this kind of model are the so-called conditional transformation models (Hothorn, 2018). Those models have the advantage of flexibly estimating the conditional distribution of the variable of interest.

There is one further model type which is very similar to quantile regression. Expectile regression is a generalization of the mean regression in the same way that quantile regression is a generalization of the median regression. The resulting regression functions minimize asymmetrically weighted squared residuals, where the weights are the same as the weights in quantile regression. The interpretation of those models is slightly harder than for quantile regression yet it is very useful in the analysis of financial risks, for example, the expected shortfall (Taylor, 2008).

5 Summary and Recommendations

Our goal in this article is to explain quantile regression in a simple way, in order to provide some guidance in the decision for which research questions it should be used. When only being interested in the mean and its characteristics and predictions, standard tools can already provide enough information. Quantile regression on the other hand is suitable for the following situations:

When being interested in events at the ‘limits of probability’: One example for this type of question was described in this article using the data from studies on stunting and obesity. Similar models were used in many publications such as in Fenske et al. (2011). In general, many medical questions might not search for answers on the centre of a distribution. Reliable results for extreme quantiles can however only be produced if large datasets are available.

If the conditional distribution does not follow a known distribution: For approaches such as GAMLSS or even glms, it is important to know the conditional distribution. Datasets hardly ever follow those distributions perfectly. This is however only important if the deviation changes the answers and if the focus is on the entire conditional distribution. In case of doubts, it makes sense to simply run both types of analyses (GAMLSS and quantile regression) and then compare the results. In many cases, graphical analyses of the data will shed light on the question for the appropriate distribution.

If there are a lot of outliers in the conditional distribution: Quantile regression is able to cope with outliers better due to its robustness. Outliers only have an influence on the quantile curves close to them.

In the case of heteroscedasticity: If the variance depends on the covariates, quantile regression can capture this effect. Just like earlier, most datasets will slightly violate the assumption of homoscedasticity and the model should be chosen based on the effect this violation has. In this case, just like for violation of the distributional assumptions, graphical analyses should be done to find out whether there is heteroscedasticity in the dataset.

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QUANTILE REGRESSION IN R: A VIGNETTE

ROGER KOENKER

ABSTRACT. Quantile regression is an evolving body of statistical methods for estimating and drawing inferences about conditional quantile functions. An implementation of these methods in the R language is available in the package `quantreg`. This vignette offers a brief tutorial introduction to the package. R and the package `quantreg` are open-source software projects and can be freely downloaded from CRAN: <http://cran.r-project.org>.

1. INTRODUCTION

Beran's (2003) provocative definition of statistics as "the study of algorithms for data analysis" elevates computational considerations to the forefront of the field. It is apparent that the evolutionary success of statistical methods is to a significant degree determined by considerations of computational convenience. As a result, design and dissemination of statistical software has become an integral part of statistical research. Algorithms are no longer the exclusive purview of the numerical analyst, or the proto-industrial software firm; they are an essential part of the artisanal research process. Fortunately, modern computing has also transformed the software development process and greatly facilitated collaborative research; the massive collective international effort represented by the R project exceeds the most idealistic Marxist imagination.

Algorithms have been a crucial part of the research challenge of quantile regression methods from their inception in the 18th century. Stigler [1984] describes an amusing episode in 1760 in which the itinerant Croatian Jesuit Rudjer Boscovich sought computational advice in London regarding his nascent method for median regression. Ironically, a fully satisfactory answer to Boscovich's questions only emerged with dawn of modern computing. The discovery of the simplex method and subsequent developments in linear programming have made quantile regression methods competitive with traditional least squares methods in terms of their computational effort. These computational developments have also played a critical role in encouraging a deeper appreciation of the statistical advantages of these methods.

Since the early 1980's I have been developing software for quantile regression: initially for the S language of Chambers and Becker (1984), later for its commercial manifestation Splus, and since 1999 for the splendid open source dialect R, initiated by Ihaka and Gentleman [1996] and sustained by the R Development Core Team [2003]. Although there is now some functionality for quantile regression in most of the major commercial statistical packages, I have a natural predilection for the R environment and the software that I have developed for R. In what follows, I have tried to provide a brief tutorial introduction to this environment for quantile regression.

2. WHAT IS A VIGNETTE?

This document was written in the Sweave format of Leisch [2003]. Sweave is an implementation designed for R of the literate programming style advocated by Knuth [1992]. The format permits a natural interplay between code written in R, the output of that code, and commentary on the code. Sweave documents are preprocessed by R to produce a $\text{\LaTeX}$ document that may then be processed by conventional methods. Many R packages now have Sweave vignettes describing their basic functionality. Examples of vignettes can be found for many of the R packages including this one for the `quantreg` packages in the source distribution directory `inst/doc`.

3. GETTING STARTED

I will not attempt to provide another introduction to R. There are already several excellent resources intended to accomplish this task. The books of Dalgaard [2002] and Venables and Ripley [2002] are particularly recommended. The CRAN website link to contributed documentation also offers excellent introductions in several languages.

R is an open source software project and can be freely downloaded from the CRAN website along with its associated documentation. For unix-based operating systems it is usual to download and build R from source, but binary versions are available for most computing platforms and can be easily installed. Once R is running the installation of additional packages is quite straightforward. To install the quantile regression package from R one simply types,

```
> install.packages("quantreg")
```

Provided that your machine has a proper internet connection and you have write permission in the appropriate system directories, the installation of the package should proceed automatically. Once the `quantreg` package is installed, it needs to be made accessible to the current R session by the command,

```
> library(quantreg)
```

These procedures provide access to an enormous variety of specialized `packages` for statistical analysis. As we proceed a variety of other packages will be called upon.

Online help facilities are available in two modalities. If you know precisely what you are looking for, and would simply like to check the details of a particular command you can, for example, try:

```
> help(package="quantreg")
> help(rq)
```

The former command gives a brief summary of the available commands in the package, and the latter requests more detailed information about a specific command. A convenient shorthand for the latter command is to type simply `?rq`. More generally one can initiate a web-browser help session with the command,

```
> help.start()
```

and navigate as desired. The browser approach is better adapted to exploratory inquiries, while the command line approach is better suited to confirmatory ones.

A valuable feature of R help files is that the examples used to illustrate commands are executable, so they can be pasted into an R session, or run as a group with a command like,

```
> example(rq)
```

The examples for the basic `rq` command include an analysis of the Brownlee stack-loss data: first the median regression, then the first quantile regression is computed, then the full quantile regression process. A curious feature of this often analysed data set, but one that is very difficult to find without quantile regression fitting, is the fact the 8 of the 21 points fall exactly on a hyperplane in 4-space.

The second example in the `rq` helpfile computes a weighted univariate median using randomly generated data. The original Engel [1857] data on the relationship between food expenditure and household income is considered in the third example. The data is plotted and then six fitted quantile regression lines are superimposed on the scatterplot. The final example illustrates the imposition of inequality constraints on the quantile regression coefficients using a simulated data set.

Let's consider the median regression results for the Engel example in somewhat more detail. Executing,

```
> data(engel)
> fit1 <- rq(foodexp ~ income, tau = .5, data = engel)
```

assigns the output of the median regression computation to the object `fit1`. In the command `rq()` there are also many options. The first argument is a “formula” that specifies the model that is desired. In this case we wanted to fit a simple bivariate linear model so the formula is just $y \sim x$, if we had two covariates we could say, e.g. $y \sim x+z$. Factor variables, that is variables taking only a few discrete values, are treated specially by the formula processing and result in a group of indicator (dummy) variables.

If we would like to see a concise summary of the result we can simply type,

```
> fit1
Call:
rq(formula = foodexp ~ income, tau = 0.5, data = engel)
```

```
Coefficients:
(Intercept)      income
 81.4822474    0.5601806
```

```
Degrees of freedom: 235 total; 233 residual
```

By convention for all the R linear model fitting routines, we see only the estimated coefficients and some information about the model being estimated. To obtain a more detailed evaluation of the fitted model, we can use,

```
> summary(fit1)
Call: rq(formula = foodexp ~ income, tau = 0.5, data = engel)
```

```
tau: [1] 0.5
```

```
Coefficients:
      coefficients lower bd  upper bd
```

```
(Intercept) 81.48225    53.25915 114.01156
income      0.56018     0.48702  0.60199
```

The resulting table gives the estimated intercept and slope in the first column and confidence intervals for these parameters in the second and third columns. By default, these confidence intervals are computed by the rank inversion method described in Koenker [2005], Section 3.4.5. To extract the residuals or the coefficients of the fitted relationship we can write,

```
> r1 <- resid(fit1)
> c1 <- coef(fit1)
```

They can then be easily used in subsequent calculations.

4. OBJECT ORIENTATION

A brief digression on the role of object orientation in R is perhaps worthwhile at this juncture. Expressions in R manipulate objects, objects may be data in the form of vectors, matrices or higher order arrays, but objects may also be functions, or more complex collections of objects. Objects have a class and this class identifier helps to recognize their special features and enables functions to act on them appropriately. Thus, for example, the function `summary` when operating on an object of class `rq` as produced by the function `rq` can act quite differently on the object than it would if the object were of another class, say `lm` indicating that it was the product of least squares fitting. Summary of a data structure like a matrix or data.frame would have yet another intent and outcome. In the earlier dialects of S and R methods for various classes were distinguished by appending the class name to the method separated by a “.”. Thus, the function `summary.rq` would summarize an `rq` object, and `summary.lm` would summarize an `lm` object. In either case the main objective of the summary is to produce some inferential evidence to accompany the point estimates of parameters. Likewise, plotting of various classes of R objects can be carried out by the expression `plot(x)` with the expectation that the plot command will recognize the class of the object `x` and proceed accordingly. More recently, Chambers [1998] has introduced an elegant elaboration of the class, method-dispatch framework for S and R.

Assignment of objects is usually accomplished by the operator `<-`, and once assigned these new objects are available for the duration of the R session, or until they are explicitly removed from the session. R is an open source language so *all* of the source files describing the functionality of the language are ultimately accessible to the individual user, and users are free to modify and extend the functionality of the language in any way they see fit. To accomplish this one needs to be able to find functions and modify them. This takes us somewhat beyond the intended tutorial scope of this vignette, however suffice it to say that most of the functions of the `quantreg` package you will find used below, can be viewed by simple typing the name of the function perhaps concatenated with a class name.

5. FORMAL INFERENCE

There are several alternative methods of conducting inference about quantile regression coefficients. As an alternative to the rank-inversion confidence intervals, one can obtain a more conventional looking table of coefficients, standard errors, t-statistics, and p-values using the `summary` function:

```
> summary(fit1, se = "nid")
Call: rq(formula = foodexp ~ income, tau = 0.5, data = engel)

tau: [1] 0.5

Coefficients:
                Value      Std. Error t value  Pr(>|t|)
(Intercept) 81.48225 19.25066    4.23270 0.00003
income       0.56018  0.02828   19.81032 0.00000
```

The standard errors reported in this table are computed as described in Section 3.2.3 for the quantile regression sandwich formula, and using the Hall-Sheather bandwidth rule. To obtain the Powell kernel version of the covariance matrix estimate, one specifies the option `se="ker"` in the `summary` command. It is also possible to control the bandwidths employed with the `bandwidth` option. Another option available in `summary.rq` is to compute bootstrapped standard errors. This is accomplished by specifying the option `se="boot"`. There are currently three flavors of the bootstrap available: the standard *xy*-pair bootstrap, the Parzen, Wei, and Ying [1994] version, and the Markov chain marginal bootstrap of He and Hu [2002] and Kocherginsky, He, and Mu [2004]. There is also the ability to specify *m* out of *n* versions of the bootstrap in which the sample size of the bootstrap samples is different from (typically smaller than) the original sample size. This “subsampling” approach has a number of advantages, not the least of which is that it can be considerably faster than the full *n* out of *n* version. By default `summary` also produces components estimating the full covariance matrix of the estimated parameters and its constituent pieces. For further details, see the documentation for `summary.rq`. In the case of the bootstrap methods the full matrix of bootstrap replications is also available.

There are several options to the basic fitting routine `rq`. An important option that controls the choice of the algorithm used in the fitting is `method`. The default is `method = "br"` which invokes a variant of the Barrodale and Roberts [1974] simplex algorithm described in Koenker and d’Orey [1987]. For problems with more than a few thousand observations it is worthwhile considering `method = "fn"` which invokes the Frisch-Newton algorithm described in Portnoy and Koenker [1997]. Rather than traversing around the exterior of the constraint set like the simplex method, the interior point approach embodied in the Frisch-Newton algorithm burrows from within the constraint set toward the exterior. Instead of taking steepest descent steps at each intersection of exterior edges, it takes Newton steps based on a log-barrier Lagrangian form of the objective function. Special forms of Frisch-Newton are available for problems that include linear inequality constraints and for problems with sparse design matrices. For extremely large problems with plausibly exchangeable observations `method = "pfn"` implements a version of the Frisch-Newton algorithm with a preprocessing step that can further speed things up considerably.

In problems of moderate size where the default simplex option is quite practical, the parametric programming approach to finding the rank inversion confidence intervals can be rather slow. In such cases it may be advantageous to try one of the other inference methods based on estimation of the asymptotic covariance matrix, or to consider the bootstrap. Both approaches are described in more detail below.

To provide a somewhat more elaborate visualization of the Engel example consider an example that superimposes several estimated conditional quantile functions on the Engel data scatterplot. In the resulting figure the median regression line appears as a solid (blue) line, and the least squares line as a dashed (red) line. The other quantile regression lines appear in grey. Note that the plotting of the fitted lines is easily accomplished by the convention that the command `abline` looks for a pair of coefficients, which if found are treated as the slope and intercept of the plotted line. There are many options that can be used to further fine tune the plot. Looping over the quantiles is also conveniently handled by R's `for` syntax.

Often it is useful to compute quantile regressions on a discrete set of τ 's; this can be accomplished by specifying `tau` as a vector in `rq`:

```
> xx <- income - mean(income)
> fit1 <- summary(rq(foodexp~xx,tau=2:98/100))
> fit2 <- summary(rq(foodexp~xx,tau=c(.05, .25, .5, .75, .95)))
```

The results can be summarized as a plot.

```
> pdf("engelcoef.pdf",width=6.5,height=3.5)
> plot(fit1,mfrow = c(1,2))
> dev.off()
```

or by producing a latex-formatted table.

```
> latex(fit2, caption="Engel's Law", transpose=TRUE)
```

The `pdf` command preceding the plot tells R that instructions for the plotting should be written in encapsulated pdf format and placed in the file `engelcoef.pdf`. Such files are then conveniently included in L^AT_EX documents, for example. The `dev.off()` command closes the current pdf device and concludes the figure. The horizontal lines in the coefficient plots represent the least squares fit and its associated confidence interval.

In the one-sample setting we know that integrating the quantile function over the entire domain $[0,1]$ yields the mean of the (sample) distribution,

$$\mu = \int_{-\infty}^{\infty} x dF(x) = \int_0^1 F^{-1}(t) dt.$$

Similarly, in the coefficient plots we may expect to see that integrating individual coefficients yields roughly mean effect as estimated by the associated least squares coefficient. One should be cautious, however, about this interpretation in very heterogeneous situations. For the Engel data, note that the least squares intercept is significantly above any of the fitted quantile regression curves in our initial scatter plot. The least squares fit is strongly affected by the two outlying observations with relatively low food expenditure; their attraction tilts the fitted line so its intercept drawn upward. In fact, the intercept for the Engel model is difficult to interpret since it asks us to consider food expenditure for households with zero income. Centering the covariate observations so they have mean zero, as we have done prior to computing `fit1` for the coefficient plot restores a reasonable interpretation of the intercept parameter. After centering the least squares estimate of the intercept is a prediction of mean food expenditure for a household with mean income, and the quantile regression intercept, $\hat{\alpha}(\tau)$ is a prediction of the τ th quantile of food expenditure for households with mean income. In the terminology of Tukey, the “intercept” has become a “centercept.”

```

> library(quantreg)
> data(engel)
> attach(engel)
> plot(income,foodexp,cex=.25,type="n",xlab="Household Income", ylab="Food Expenditure")
> points(income,foodexp,cex=.5,col="blue")
> abline(rq(foodexp~income,tau=.5),col="blue")
> abline(lm(foodexp~income),lty=2,col="red") #the dreaded ols line
> taus <- c(.05,.1,.25,.75,.90,.95)
> for( i in 1:length(taus)){
+     abline(rq(foodexp~income,tau=taus[i]),col="gray")
+ }

```

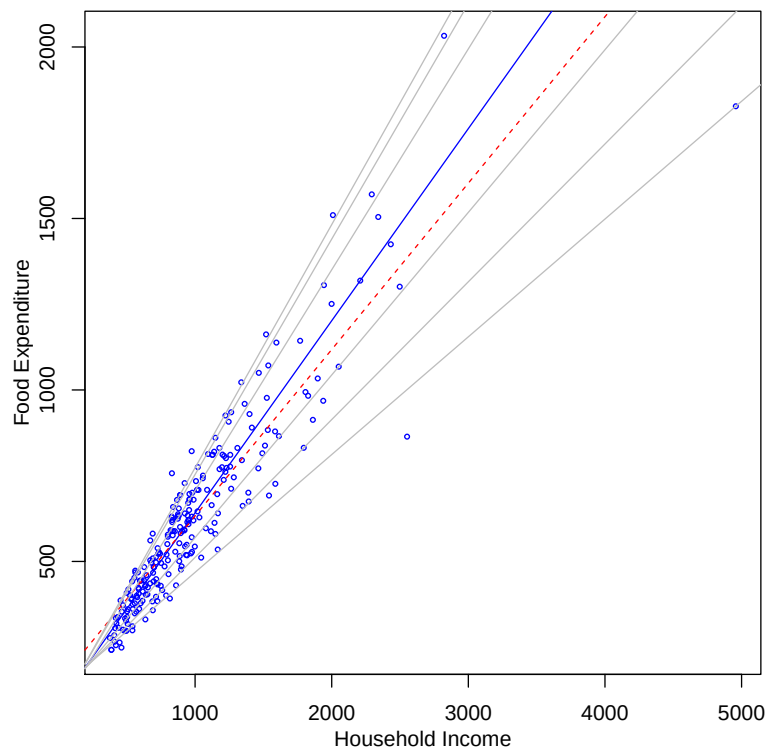


FIGURE 1. Scatterplot and Quantile Regression Fit of the Engel Food Expenditure Data: The plot shows a scatterplot of the Engel data on food expenditure vs household income for a sample of 235 19th century working class Belgian households. Superimposed on the plot are the $\{.05, .1, .25, .75, .90, .95\}$ quantile regression lines in gray, the median fit in solid black, and the least squares estimate of the conditional mean function as the dashed (red) line.

The `latex` command produces a $\text{\LaTeX}$ formatted table that can be easily included in documents. In many instances the plotted form of the results will provide a more economical and informative display. It should again be stressed that since the quantile regression functions and indeed all of R is open source, users can

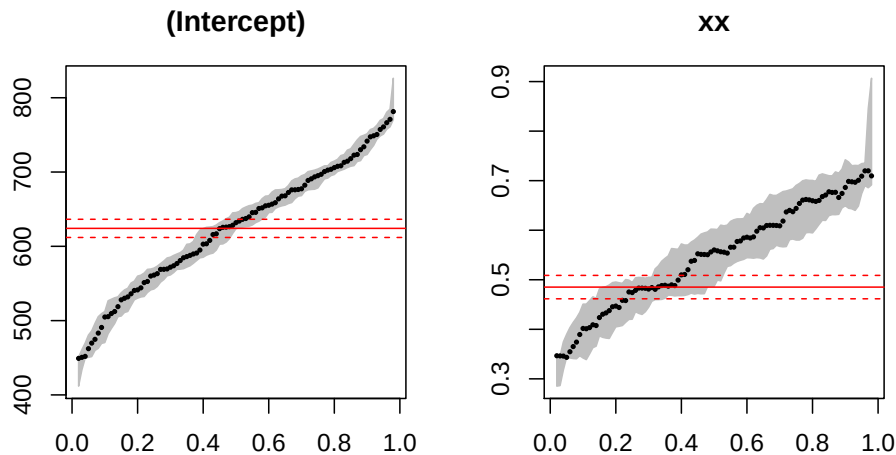


FIGURE 2. Engel Coefficient Plots: the slope and intercept of the estimated linear quantile regression linear for the Engel food expenditure data are plotted as a function of τ . Note that the household income variable has been centered at its mean value for this plot, so the intercept is really a centercept and estimates the quantile function of food expenditure conditional on mean income.

TABLE 1. Engel's Law

Quantiles	(Intercept)	xx
0.05	462.223 (450.572, 480.503)	0.343 (0.343, 0.390)
0.25	561.277 (542.572, 570.726)	0.474 (0.420, 0.494)
0.50	631.845 (623.706, 645.461)	0.560 (0.487, 0.602)
0.75	695.123 (678.663, 704.105)	0.644 (0.580, 0.690)
0.95	760.745 (751.092, 771.916)	0.709 (0.674, 0.734)

always modify the available functions to achieve special effects required for a particular application. When such modifications appear to be of general applicability, it is desirable to communicate them to the package author, so they could be shared with the larger community.

If we want to see *all* the distinct quantile regression solutions for a particular model application we can specify a tau outside the range $[0,1]$, e.g.

```
> z <- rq(foodexp~income, tau=-1)
```

This form of the function carries out the parametric programming steps required to find the entire sample path of the quantile regression process. The returned object is of class `rq.process` and has several components: the primal solution in `z$sol`, and the dual solution in `z$dsol`. In interactive mode typing the name of

an R object causes the program to print the object in some reasonably intelligible manner determined by the print method designated for the object's class. Again, plotting is often a more informative means of display and so there is a special `plot` method for objects of class `rq.process`.

Estimating the conditional quantile functions of y at a specific values of x is also quite easy. In the following code we plot the estimated empirical quantile functions of food expenditure for households that are at the 10th percentile of the sample income distribution, and the 90th percentile. In the right panel we plot corresponding density estimates for the two groups. The density estimates employ the adaptive kernel method proposed by Silverman [1986] and implemented in the `quantreg` function `akj`. This function is particularly convenient since it permits unequal mass to be associated with the observations such as those produced by the quantile regression process.

Thus far we have only considered Engel functions that are linear in form, and the scatterplot as well as the formal testing has revealed a strong tendency for the dispersion of food expenditure to increase with household income. This is a particularly common form of heteroscedasticity. If one looks more carefully at the fitting, one sees interesting departures from symmetry that would not be likely to be revealed by the typical textbook testing for heteroscedasticity. One common remedy for symptoms like these would be to reformulate the model in log linear terms. It is interesting to compare what happens after the log transformation with what we have already seen.

Note that the flag `log="xy"` produces a plot with log-log axes, and for convenience of axis labeling these logarithms are base 10, so the subsequent fitting is also specified as base 10 logs for plotting purposes, even though base 10 logarithms are *unnatural* and would never be used in reporting numerical results. This looks much more like a classical iid error regression model, although again some departure from symmetry is visible. An interesting exercise would be to conduct some formal testing for departures from the iid assumption of the type already considered above.

6. MORE ON TESTING

Now let's consider some other forms of formal testing. A natural first question is: do the estimated quantile regression relationships conform to the location shift hypothesis that assumes that all of the conditional quantile functions have the same slope parameters. To begin, suppose we just estimate the quartile fits for the Engel data and look at the default output:

```
> fit1 <- rq(foodexp~income,tau=.25)
> fit2 <- rq(foodexp~income,tau=.50)
> fit3 <- rq(foodexp~income,tau=.75)
```

Recall that `rq` just produces coefficient estimates and `summary` is needed to evaluate the precision of the estimates. This is fine for judging whether covariates are significant at particular quantiles but suppose that we wanted to test that the slopes were the same at the three quartiles? This is done with the `anova` command as follows:

```
> anova(fit1, fit2, fit3)
```

Quantile Regression Analysis of Deviance Table

```

> x.poor <- quantile(income,.1) #Poor is defined as at the .1 quantile of the sample distn
> x.rich <- quantile(income,.9) #Rich is defined as at the .9 quantile of the sample distn
> ps <- z$sol[1,]
> qs.poor <- c(c(1,x.poor)%*%z$sol[4:5,])
> qs.rich <- c(c(1,x.rich)%*%z$sol[4:5,])
> #now plot the two quantile functions to compare
> par(mfrow = c(1,2))
> plot(c(ps,ps),c(qs.poor,qs.rich), type="n",
+      xlab = expression(tau), ylab = "quantile")
> plot(stepfun(ps,c(qs.poor[1],qs.poor)), do.points=FALSE, add=TRUE)
> plot(stepfun(ps,c(qs.poor[1],qs.rich)), do.points=FALSE, add=TRUE,
+      col.hor = "gray", col.vert = "gray")
> ## now plot associated conditional density estimates
> ## weights from ps (process)
> ps.wts <- (c(0,diff(ps)) + c(diff(ps),0)) / 2
> ap <- akj(qs.poor, z=qs.poor, p = ps.wts)
> ar <- akj(qs.rich, z=qs.rich, p = ps.wts)
> plot(c(qs.poor,qs.rich),c(ap$dens,ar$dens),type="n",
+      xlab= "Food Expenditure", ylab= "Density")
> lines(qs.rich, ar$dens, col="gray")
> lines(qs.poor, ap$dens, col="black")
> legend("topright", c("poor","rich"), lty = c(1,1), col=c("black","gray"))

```

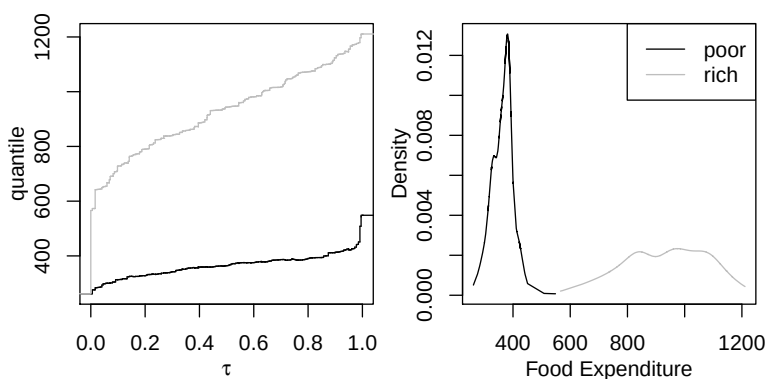


FIGURE 3. Estimated conditional quantile and density functions for food expenditure based on the Engel data: Two estimates are presented one for relatively poor households with income of 504.5 Belgian francs, and the other for relatively affluent households with 1538.99 Belgian francs.

Model: foodexp ~ income

Joint Test of Equality of Slopes: tau in { 0.25 0.5 0.75 }

	Df	Resid	Df	F value	Pr(>F)
1	2	703	15.557	2.449e-07	***

```

> plot(income,foodexp,log="xy",xlab="Household Income", ylab="Food Expenditure")
> taus <- c(.05,.1,.25,.75,.90,.95)
> abline(rq(log10(foodexp)~log10(income),tau=.5),col="blue")
> abline(lm(log10(foodexp)~log10(income)),lty = 3,col="red")
> for( i in 1:length(taus)){
+   abline(rq(log10(foodexp)~log10(income),tau=taus[i]),col="gray")
+ }

```

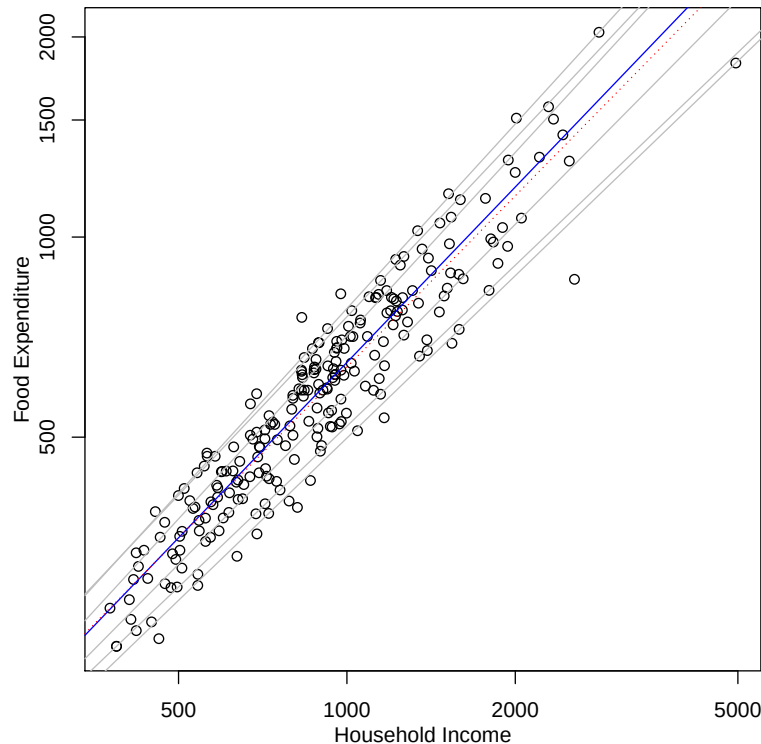


FIGURE 4. Quantile regression estimates for a log-linear version of the Engel food expenditure model.

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

This is an example of a general class of tests proposed in Koenker and Bassett [1982]. It may be instructive to look at the code for the command `anova.rq` to see how this test is carried out. The Wald approach is used and the asymptotic covariance matrix is estimated using the approach of Hendricks and Koenker [1991]. It also illustrates a general syntax for testing in R adapted to the present situation.

If you have estimated two models with different covariate specifications, but the same τ then `anova(f0,f1)` tests whether the more restricted model is preferred. Note that this requires that the models with fits say `f0` and `f1` are nested. The procedure `anova.rq` attempts to check whether the fitted models are nested, but this is not foolproof. These tests can be carried out either as Wald tests based on the estimated joint asymptotic covariance matrix of the coefficients, or using the

rank test approach described in Gutenbrunner, Jurečková, Koenker, and Portnoy [1993]. A variety of other options are described in the documentation of the function `anova.rq`.

7. INFERENCE ON THE QUANTILE REGRESSION PROCESS

In least squares estimation of linear models it is implicitly assumed that we are able to model the effects of the covariates as a pure location shift, or somewhat more generally as a location and scale shift of the response distribution. In the simplest case of a single binary treatment effect this amounts to assuming that the treatment and control distributions differ by a location shift, or a location-scale shift. Tests of these hypotheses in the two sample model can be conducted using the conventional two-sample Kolmogorov Smirnov statistic, but the appearance of unknown nuisance parameters greatly complicates the limiting distribution theory. Similar problems persist in the extension of these tests to the general quantile regression setting. Using an approach introduced by Khmaladze [1981], Koenker and Xiao [2002] consider general forms of such tests. The tests can be viewed as a generalization of the simple tests of equality of slopes across quantiles described in the previous section.

In this section we briefly describe how to implement these tests in R. The application considered is the quantile autoregression (QAR) model for weekly U.S. gasoline prices considered in Koenker and Xiao [2006]. The data consists of 699 weekly observations running from August 1990 to February, 2004. The model considered is a QAR(4) model utilizing four lags. We are interested in testing whether the classical location-shift AR(4) is a plausible alternative to the QAR(4) specification, that is whether the four QAR lag coefficients are constant with respect to τ .

To carry out the test we can either compute the test using the full quantile regression process or on a moderately fine grid of τ s:

```
> data(gasprice)
> x <- gasprice
> n <- length(x)
> p <- 5 # lag length
> X <- cbind(x[(p-1):(n-1)], x[(p-2):(n-2)], x[(p-3):(n-3)], x[(p-4):(n-4)])
> y <- x[p:n]
> T1 <- KhmaladzeTest(y ~ X,taus = 1:99/100, nullH="location")
> T2 <- KhmaladzeTest(y ~ X,taus = 10:290/300, nullH="location",se="ker")
```

When `taus` contains elements outside of the interval $(0,1)$ then the process is standardized by a simple form of the covariance matrix that assumes iid error. In the second version of the test Powell's kernel form of the sandwich formula estimate is used, see `summary.rq`. The function `KhmaladzeTest` computes both a joint test that *all* the covariate effects satisfy the null hypothesis, and a coefficient by coefficient version of the test. In this example the former component, `T1$Tn` is 5.09. This test has a 1 percent critical value of 5.56, so the test weakly rejects the null. For the Powell form the standardization the corresponding test statistic is more decisive, taking the value 4.27.

Tests of the location-scale shift form of the null hypothesis can be easily done by making the appropriate change in the `nullH` argument of the function.

8. NONLINEAR QUANTILE REGRESSION

Quantile regression models with response functions that are nonlinear in parameters can be estimated with the function `nlrq`. For such models the specification of the model formula is somewhat more esoteric than for ordinary linear models, but follows the conventions of the R command `nls` for nonlinear least squares estimation.

To illustrate the use of `nlrq` consider the problem of estimating the quantile functions of the Frank copula model introduced in Koenker [2005], Section 8.4. We begin by setting some parameters and generating data from the Frank model:

```
> n <- 200
> df <- 8
> delta <- 8
> set.seed(4003)
> x <- sort(rt(n,df))
> u <- runif(n)
> v <- -log(1-(1-exp(-delta))/(1+exp(-delta*pt(x,df))*((1/u)-1)))/delta
> y <- qt(v,df)
```

We plot the observations, superimpose three conditional quantile functions, and then estimate the same three quantile functions and plot their estimated curves as the dashed curves.

9. NONPARAMETRIC QUANTILE REGRESSION

Nonparametric quantile regression is initially most easily considered within a locally polynomial framework. Locally linear fitting is carried out by the following function:

```
> "lprq" <-
+ function(x, y, h, m=50, tau=.5)
+ {
+   xx <- seq(min(x),max(x),length=m)
+   fv <- xx
+   dv <- xx
+   for(i in 1:length(xx)) {
+     z <- x - xx[i]
+     wx <- dnorm(z/h)
+     r <- rq(y~z, weights=wx, tau=tau, ci=FALSE)
+     fv[i] <- r$coef[1.]
+     dv[i] <- r$coef[2.]
+   }
+   list(xx = xx, fv = fv, dv = dv)
+ }
```

If you study the function a bit you will see that it is simply a matter of computing a quantile regression fit at each of m equally spaced x -values distributed over the support of the observed x points. The function value estimates are returned as `fv` and the first derivative estimates at the m points are returned as `dv`. As usual you can specify τ , but now you also need to specify a bandwidth `h`.

Let's begin by exploring the effect of the `h` argument for fitting the motorcycle data.

```

> plot(x,y,col="blue",cex = .25)
> us <- c(.25,.5,.75)
> for(i in 1:length(us)){
+   u <- us[i]
+   v <- -log(1-(1-exp(-delta))/
+   (1+exp(-delta*pt(x,df))*((1/u)-1)))/delta
+   lines(x,qt(v,df))
+ }
> Dat <- NULL
> Dat$x <- x
> Dat$y <- y
> deltas <- matrix(0,3,length(us))
> FrankModel <- function(x,delta,mu,sigma,df,tau){
+   z <- qt(-log(1-(1-exp(-delta))/
+   (1+exp(-delta*pt(x,df))*((1/tau)-1)))/delta,df)
+   mu + sigma*z
+ }
> for(i in 1:length(us)){
+   tau = us[i]
+   fit <- nlrq(y~FrankModel(x,delta,mu,sigma,df=8,tau=tau),
+   data=Dat,tau= tau, start=list(delta=5,
+   mu = 0, sigma = 1),trace=TRUE)
+   lines(x, predict(fit, newdata=x), lty=2, col="green")
+   deltas[i,] <- coef(fit)
+ }

```

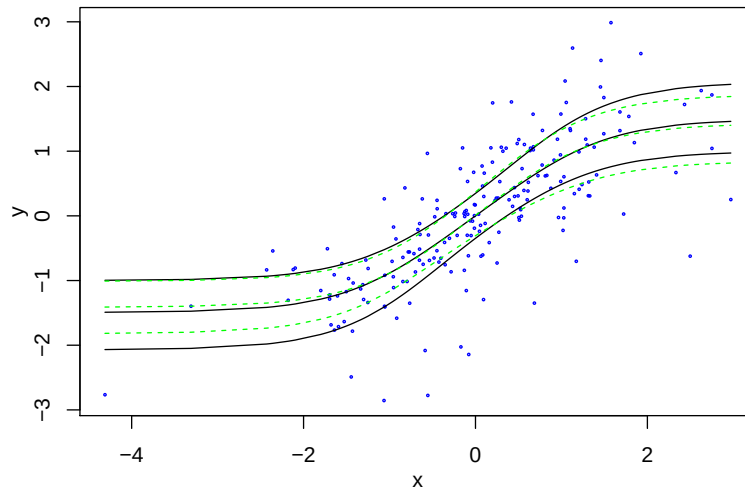


FIGURE 5. Nonlinear Conditional Quantile Estimation of the Frank Copula Model: The solid curves are the true conditional quantile functions and the corresponding estimated curves are indicated by the dashed curves.

```

> library(MASS)
> data(mcycle)
> attach(mcycle)
> plot(times, accel, xlab = "milliseconds", ylab = "acceleration")
> hs <- c(1,2,3,4)
> for(i in hs){
+   h = hs[i]
+   fit <- lprq(times, accel, h=h, tau=.5)
+   lines(fit$xx, fit$f, lty=i)
+ }
> legend(45, -70, c("h=1", "h=2", "h=3", "h=4"), lty=1:length(hs))

```

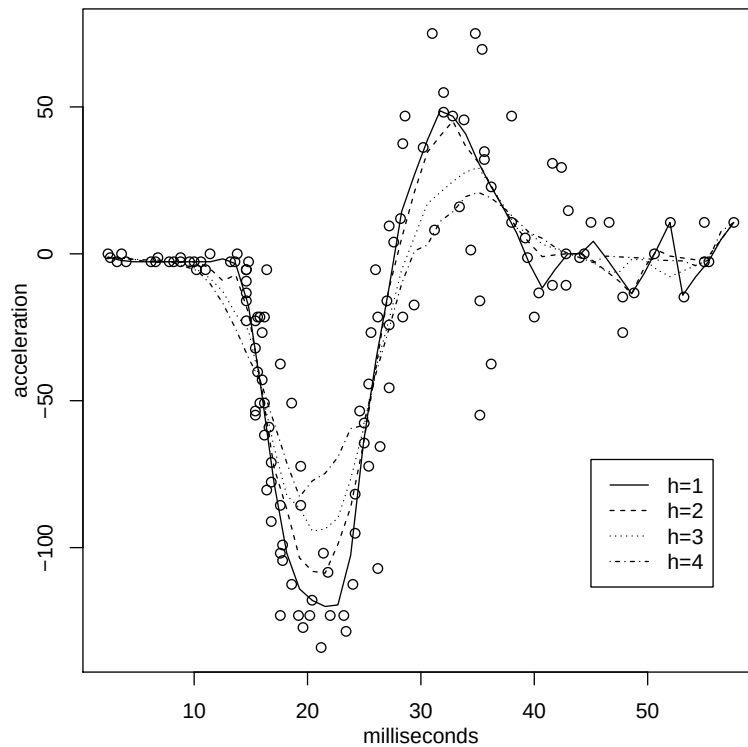


FIGURE 6. Estimation of a locally linear median regression model for the motorcycle data: four distinct bandwidths

Fitting derivatives, of course, requires somewhat larger bandwidth and larger sample size to achieve the same precision as function fitting. It is a straightforward exercise to adapt the function `lprq` so that it does locally quadratic rather than locally linear fitting.

Another simple, yet quite general, strategy for nonparametric quantile regression uses regression splines. The function `bs()` in the package `splines` gives a very flexible way to construct B-spline basis expansions. For example you can fit a new motorcycle model like this:

```

> library(splines)
> plot(times, accel, xlab = "milliseconds", ylab = "acceleration", type = "n")
> points(times, accel, cex = .75)
> X <- model.matrix(accel ~ bs(times, df=15))
> for(tau in 1:3/4){
+   fit <- rq(accel ~ bs(times, df=15), tau=tau, data=mcycle)
+   accel.fit <- X %*% fit$coef
+   lines(times, accel.fit)
+ }

```

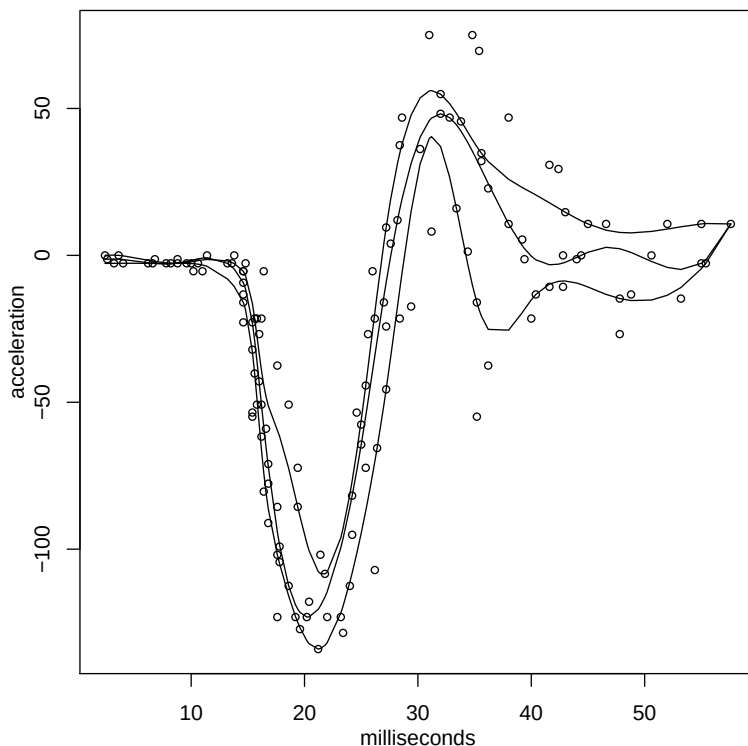


FIGURE 7. B-spline estimates of the three conditional quantile functions of the motorcycle data

The fitted conditional quantile functions do reasonably well except in the region beyond 50 milliseconds where the data is so sparse that all the quantile curves want to coalesce.

This procedure fits a piecewise cubic polynomial with 15 knots (breakpoints in the third derivative) arranged at quantiles of the x 's. (You can also explicitly specify the knot sequence and the order of the spline using the optional arguments to `bs`.) In this instance we have estimated three quantile curves for the B-spline model; a salient feature of this example is the quite dramatic variability over the time scale. There is essentially no variability for the first few milliseconds, but quite substantial variability after the crash. One advantage of the B-spline approach is that it is very easy to add a partially linear model component. If there

were another covariate, say z , it could be added as a parametric component using the usual `formula` syntax,

```
> fit <- rq(y ~ bs(x,df=5)+z,tau=.33)
```

Another appealing approach to nonparametric smoothing involves penalty methods. Koenker, Ng, and Portnoy [1994] describe total variation penalty methods for fitting univariate functions; Koenker and Mizera [2004], extend to approach to bivariate function estimation. Again, partially linear models are easily adapted, and there are also easy ways to impose monotonicity and convexity on the fitted functions. In some applications it is desirable to consider models that involve nonparametric fitting with several covariates. A tractable approach that has been explored by many authors is to build additive models. This approach is available in R for least squares fitting of smoothing spline components in the `mgcv` and `gss` packages. A prototype package for quantile regression models of this type is available using the `rqss` function of the `quantreg` package.

The function `rqss` offers a formula interface to nonparametric quantile regression fitting with total variation roughness penalties. Consider the running speed of mammals example from Chapter 7. The objective is to estimate a model for the upper envelope of the scatterplot, a model that would reflect best evolutionary practice in mammalian ambulatory efficiency. In contrast to the least squares analysis of Chappell [1989] where they are omitted, no special allowance is made for the “specials” indicated by the plotting character `s` or “hoppers” indicated by `h`. The data is plotted on a (natural) log scale, the model is fit using $\lambda = 1$ as the penalty parameter, and the fitted curve is plotted using a special plotting function that understands the structure of the objects returned from `rqss`. The estimated turning point of the piecewise linear fitted function occurs at a weight of about 40 Kg.

```
> data(Mammals)
> attach(Mammals)
```

Bivariate nonparametric fitting using the triogram methods described in Chapter 7 can be handled in a similar manner. If we consider the Cobar mining data from Green and Silverman [1994]

The `qss` term in this case requires both x and y components. In addition one needs to specify a smoothing parameter λ , and the parameter `ndum` may be used to specify the number of artificial vertices introduced into the fitting procedure in addition to the actual observations. These artificial vertices contribute to the penalty term, but not to the fidelity.

By default the fit is rendered as a contour plot, but there are also two forms of perspective plots. A conventional R `persp` plot can be obtained by passing option `render = "persp"` to the plot command. More adventurous R gonauts are encouraged to explore the option `render = "rgl"`, which produces a perspective plot in the dynamic graphics interface to the open GL library provided by the package `rgl`. Of course this package must be installed. A demonstration of how to create animations of `rqss` triogram output using `rgl` is available by running the command `demo(cobar)`.

Another advantage of the penalty approach embodied in `rqss` is that it is straightforward to impose additional qualitative constraints on the fitted functions. Univariate functions can be constrained to be monotone and/or convex or concave.

```

> x <- log(weight)
> y <- log(speed)
> plot(x,y, xlab="Weight in log(Kg)", ylab="Speed in log(Km/hour)", type="n")
> points(x[hoppers],y[hoppers],pch = "h", col="red")
> points(x[specials],y[specials],pch = "s", col="blue")
> others <- (!hoppers & !specials)
> points(x[others],y[others], col="black",cex = .75)
> fit <- rqss(y ~ qss(x, lambda = 1),tau = .9)
> plot(fit, add = TRUE)

```

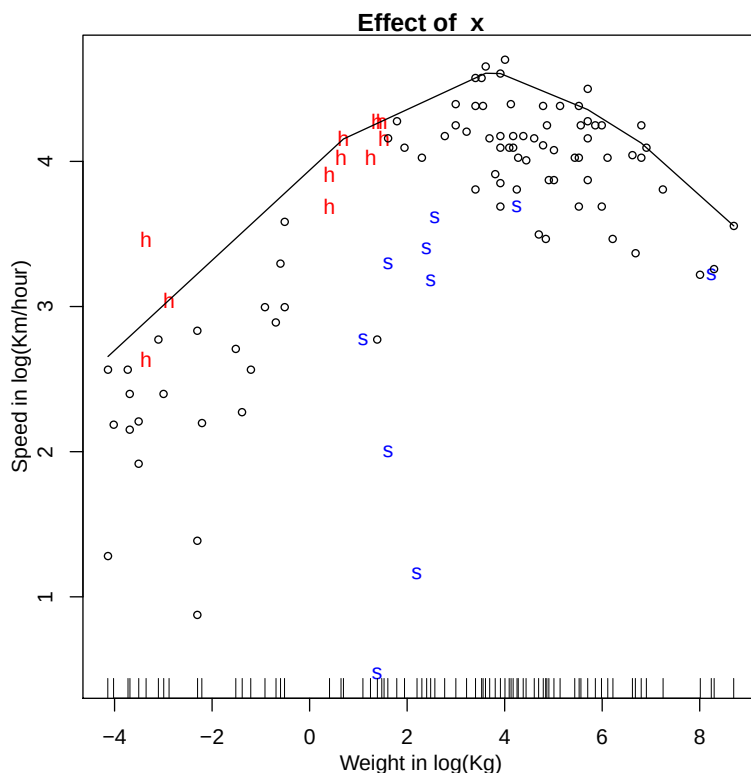


FIGURE 8. Maximal Running Speed of Terrestrial Mammals: The figure illustrates the relationship between adult body mass and maximal running speed for 107 species of terrestrial mammals. The piecewise linear curve is an estimate of the .90 conditional quantile function estimated subject to a constraint on the total variation of the function gradient.

Bivariate functions can be constrained to be either convex or concave. This functionality is implemented by simply imposing nonnegativity constraints on certain linear combinations of model parameters and illustrates one of many possible applications for such constraints. An interesting open research problem involves formal inference on such constraints.

```

> data(CobarOre)
> fit <- rqss(z ~ qss(cbind(x,y), lambda = .01, ndum=100), data = CobarOre)
> plot(fit, axes = FALSE, xlab = "", ylab = "")
> rm(list=ls())

```

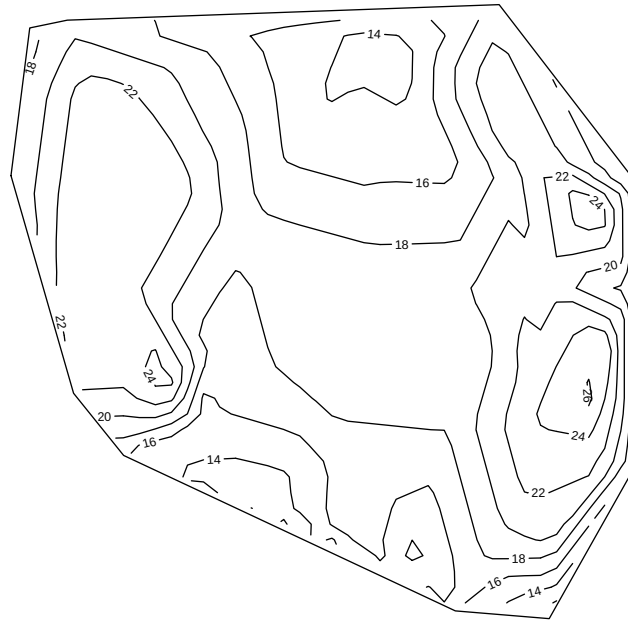


FIGURE 9. Contour plot of a triogram fit of the Cobar mining data.

10. CONCLUSION

A few of the capabilities of the R **quantreg** package have been described. Inevitably, new applications will demand new features and reveal unforeseen bugs. In either case I hope that users will send me their comments and suggestions. This document will be periodically updated and the current version will be made available in the R distribution of **quantreg**. See the R command **vignette()** for details on how to find and view vignettes from within R.

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Quantile regression for partially linear varying-coefficient model with censoring indicators missing at random

Yu Shen, Han-Ying Liang  

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Abstract

In this paper, we focus on the partially linear varying-coefficient quantile regression model when the data are right censored and the censoring indicator is missing at random. Based on the calibration and imputation methods, a three-stage approach is proposed to construct the estimators of the linear part and the nonparametric varying-coefficient function for this model. At the same time, we discuss the variable selection of the covariates in the linear part by adopting adaptive LASSO penalty. Under appropriate assumptions, the asymptotic normality of the proposed estimators is established, and the penalized estimators are proven to have the oracle property. Simulation study and a real data analysis are conducted to evaluate the performance of the proposed estimators.

Introduction

Flexible and refined statistical models are widely sought in a large array of statistical problems. Partially linear varying-coefficient (PLVC) model, as a combination of the partially linear model and varying-coefficient model, has drawn much attention of many researchers, and various methods have been proposed to estimate its parametric part and nonparametric function. See Zhang et al. (2002), Ahmad et al. (2005), Lam and Fan (2008), among others.

Most existing estimation methods for the PLVC model focus on mean regression by least squares or likelihood method. However they are sensitive to outliers and may be inefficient for many non-normal errors. The quantile regression (QR) introduced by Koenker and Bassett (1978) is more robust in exploring the underlying relationship between the covariates and the response. We refer to Koenker (2005) for a comprehensive overview of the QR. Some authors have discussed the PLVC QR model. Wang et al. (2009) used B-spline basis functions to define estimator of the QR with longitudinal data; Kai et al. (2011) proposed a novel three-stage estimation procedure and studied variable selection; Cai and Xiao (2012) discussed the QR in dynamic models with time series.

In many applications, especially in survival analysis and biomedical studies, the response cannot be completely observed due to possible right censoring, such as withdrawal of patients from the study, or death from a cause unrelated to the disease of being studied. Based on the linear model, the censored QR was first studied by Powell (1986) for the case of fixed censoring that assumes known censoring time for all observations. For the random right-censored model, Ying et al. (1995) proposed a semi-parametric procedure for median regression model, Portnoy (2003) developed a recursively reweighted estimation of the QR by the classical Kaplan–Meier estimator, Wang and Wang (2009) proposed a locally weighted censored QR approach that adopts the redistribution of mass idea and employs a local reweighting scheme, and Leng and Tong (2013) proposed a QR estimator through an unbiased estimating equation. Based on the varying-coefficient model, Yin et al. (2014) proposed a QR model and estimated regression parameters by generalized method of moments, and Xie et al. (2015) adopted a weighted inverse probability approach to develop the QR estimation. Furthermore, the censoring indicator may not be observed completely in practice. For instance, as described by Wang and Shen (2008), in clinical trials, individuals may fail from one of multiple causes, one of which is of interest. The time to death from the cause of interest may be censored by a death from a different cause. However, cause of death may sometimes be unavailable, for example: whether the death is attributable to the cause of interest may require information that is not collected or lost, or it may be difficult to determine the cause for some patients. In such cases, some censoring indicators are missing. In this paper, we assume that the censoring indicator is missing at random (MAR). The MAR assumption is common in statistical analysis involving missing data and is reasonable in many practical situations. See Little and Rubin (1987). Some authors have discussed the QR with random missing data. Yi and He (2009) investigated the linear QR model with missing response, and Sherwood et al. (2013) and Yang and Liu (2014) studied the linear QR model with missing covariates. For the varying-coefficient model with missing covariates, Sun and Sun (2015) studied the QR, Tang and Zhou (2015) studied the composite QR. For the right-censored data with missing censoring indicator, Subramanian (2004) and Wang and Ng (2008) suggested different estimators of the survival function, Song et al. (2010) and Qiu et al. (2015) discussed additive hazards regression model, and Wang and Dinse (2011) and Li and Wang (2012) proposed weighted least squared estimators for linear mean regression model. However there is no literature studying the PLVC QR model with censoring indicator MAR so far.

In addition, it is well known that variable selection for predictors has drawn the attention of many researchers. Various powerful penalization methods have been developed for the variable selection, such as LASSO (Tibshirani, 1996), SCAD (Fan and Li, 2001), adaptive LASSO (Zou, 2006), and so on. Recently, the effective variable selection procedure has been developed for the PLVC model, for instance, under the complete data setting, Li and Liang (2008) proposed the nonconcave penalized quasi-likelihood method for variable selection in the PLVC mean regression model, and Kai et al. (2011) discussed the variable selection for the PLVC composite QR model. However there is little research for the variable selection of the PLVC QR model related to the right censored data with censoring indicator MAR.

In this paper, we focus on the PLVC QR model when the data are right censored and the censoring indicator is MAR. Using calibration and imputation methods, we suggest a three-stage approach to estimate both parameters of the linear part and the functions of the nonparametric varying-coefficient part in the model. Furthermore, when the covariates in the linear part are high-dimensional, we construct the penalized estimators by applying adaptive LASSO penalty, and study the variable selection of the model. A simulation study and a real data analysis are conducted to evaluate the performance of the proposed estimators.

The rest of the paper is organized as follows. In Section 2, a three-stage approach is proposed to construct the estimators in the PLVC QR model with the censoring indicator MAR, and the penalized estimators of the parameter in the linear part by applying adaptive LASSO penalty are also discussed. Main results are described in Section 3. Numerical study is presented in Section 4. The proofs of main results are given in Section 5.

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Section snippets

Methodology

Consider the following partially linear varying-coefficient QR model

$T_i = X_i^T \beta_\tau + Z_i^T \alpha_\tau(U_i) + \varepsilon_i$, $i = 1, 2, \dots, n$, where T_i is the response variable, $X_i \in \mathbb{R}^p$, $Z_i \in \mathbb{R}^q$ and $U_i \in \mathbb{R}$ are the explanatory variables, $\beta_\tau = (b_{\tau 1}, b_{\tau 2}, \dots, b_{\tau p})^T$ is an unknown p -dimensional parameter vector, $\alpha_\tau(\cdot) = (\alpha_{\tau 1}(\cdot), \alpha_{\tau 2}(\cdot), \dots, \alpha_{\tau q}(\cdot))^T$ is an unknown q -dimensional coefficient function vector, and ε_i is the random error whose τ th quantile on

$W_i = (X_i, Z_i, U_i)$ equals zero for $\tau \in (0, 1)$. Let $Q_{T_i}(\tau|W_i)$ be the condition quantile of T_i given W_i , then $Q_{T_i}(\tau|W \dots$

Main results

In the sequel, let C denote generic finite positive constant, whose value may be different from line to line. We first give some notations.

- (1) $H(\cdot)$ is the distribution function of Y ; $F(\cdot|W)$ and $f(\cdot|W)$ are the conditional distribution function and the conditional density function of ε given W , respectively; $f_U(\cdot)$ is the density function of U . Let $\pi(y) = E(\xi|Y = y)$ and $D_U = \{u : f_U(u) \neq 0\}$
- (2) Let $\nabla m_0(Y, W, \theta) = \left(\frac{\partial m_0(Y, W, \theta)}{\partial \theta_1}, \dots, \frac{\partial m_0(Y, W, \theta)}{\partial \theta_t} \right)^T$ and $I(\theta) = E \left\{ \frac{\xi \nabla m_0(Y, W, \theta) \nabla^T m_0(Y, W, \theta)}{m} \right\} \dots$

...

Simulation study

In this section, we study the finite-sample performance of the proposed estimators through Monte Carlo simulation. We choose $K(\cdot)$ and $L(\cdot)$ to be the Epanechnikov kernel

$K(x) = L(x) = \frac{3}{4}(1 - x^2)_+$, set the number of Monte Carlo simulation $M = 200$ and take $\tau = 0.5$. We compare the performance of the estimators based on the calibration method (CA), imputation method (IM) and CC method, where the CC method (complete case analysis), which ignores the missing data and gets the estimators by only using the data ...

Proofs of main results

Lemma 5.1

Under Assumptions (A7)–(A8), we have $\widehat{G}(y) - G(y) = \frac{1-G(y)}{n} \sum_{j=1}^n \psi_j(y) + o_p(n^{-1/2})$, where $\psi_j(y) = \frac{-[\xi_j - \pi(y_j)][\delta_j - \mu(Y_j)]I(Y_j \leq y)}{\pi(y_j)[1-H(Y_j)]} + \int_0^{Y_j \wedge y} \frac{dH_0(s)}{[1-H(s)]^2} + \frac{I(Y_j \leq y, \delta_j=0)}{1-H(Y_j)}$ and $H_0(t) = P(Y_j > t, \delta_j = 0)$.

...

The result in Lemma 5.1 can be found in Li and Wang (2012) (or see their (B.15)). In addition, from their proof, we have $E[\psi_j(Y_i)] = 0$ for $i \neq j$.

Proof of Theorem 3.1

We prove only the result about $(\tilde{\alpha}_\tau^2(u_0), \tilde{\beta}_\tau^2)$, the result related to $(\tilde{\alpha}_\tau^1(u_0), \tilde{\beta}_\tau^1)$ can be proved similarly. ...

...

Acknowledgments

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...In recent years, some authors have done a little work on quantile regression when some censoring indicators are missing. For example, Shen and Liang [33] considered the data with the responses being right-censored, while the censoring indicator is missing at random (MAR) in the partially linear varying-coefficient quantile regression model. Zou et al. [51] studied quantile regression and variable selection for the partially linear single-index model with missing censoring indicators....

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...Sherwood et al. (2013) and Yang and Liu (2016) investigated the linear QR models with missing covariates. For the right-censored data with missing censoring indicators, Shen and Liang (2017) proposed a three-stage approach to construct the estimators of the linear part and nonparametric function for partially linear varying-coefficient QR models. As far as we know, there is no literature related to the PLSI model with censoring indicators MAR, and we shall focus on the QR estimation problems for this model....

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